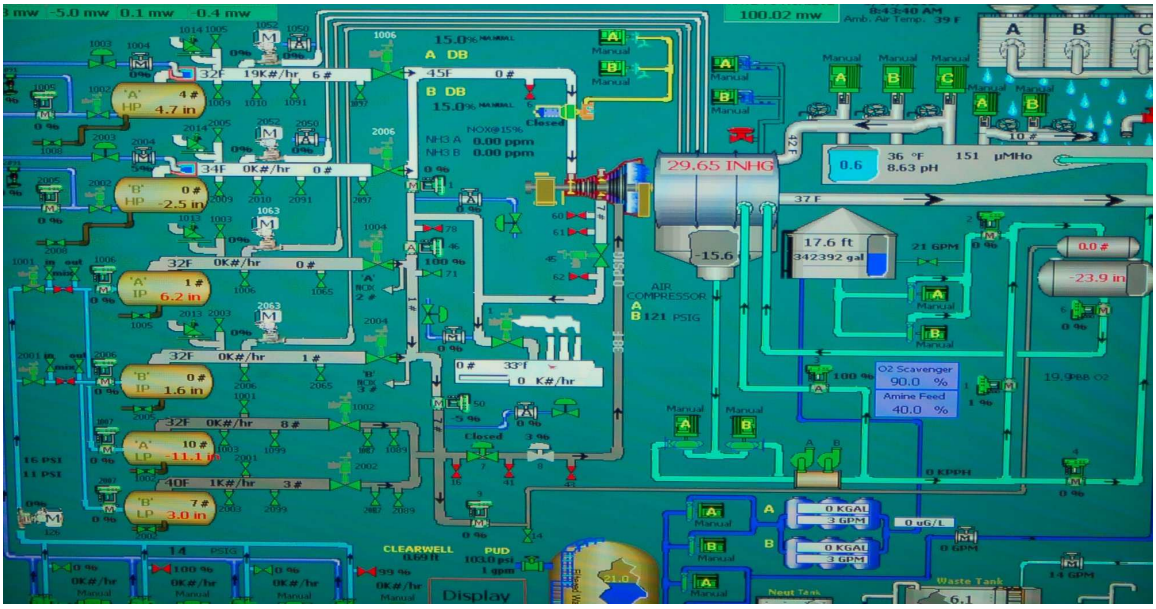
Lessons In Industrial Instrumentation



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Version 2.33 (development) – Last update 9 October 2024

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- Version 0.8 – released September 8, 2009 for public use (SafeCreative registration code `0909094400980`)
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- Version 1.6 – released May 4, 2010 for public use (SafeCreative registration code `1005046194859`)
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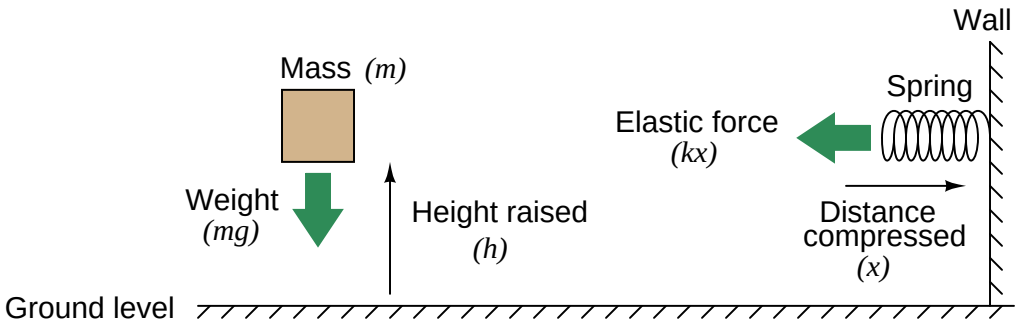
[1] Version numbers ending in odd digits are developmental (e.g. 0.7, 1.23, 4.5), with only the latest revision made accessible to the public. Version numbers ending in even digits (e.g. 0.6, 1.0, 2.14) are considered "public-release" and will be archived. Version numbers beginning with zero (e.g. 0.1, 0.2, etc.) represent early editions that were substantially incomplete.

Chapter 4

DC electricity

4.1 Electrical voltage

Voltage is the amount of *specific potential energy* available between two points in an electric circuit. Potential energy is energy that is potentially available to do work. Looking at this from a classical physics perspective, potential energy is what we accumulate when we lift a weight above ground level, or when we compress a spring:



In either case, potential energy is calculated by the work done in exerting a force over a parallel distance. In the case of the weight, potential energy (E_p) is the simple product of weight (gravity g acting on the mass m) and height (h):

$$E_p = mgh$$

Energy calculations for springs are more complex than for weights. The force exerted by the spring against the compressing motion increases with compression ($F = kx$, where k is the elastic constant of the spring). It does not remain steady as the force of weight does for the lifted mass. Therefore, the potential energy equation is nonlinear:

$$E_p = \frac{1}{2}kx^2$$

Releasing the potential energy stored in these mechanical systems is as simple as dropping the mass, or letting go of the spring. The potential energy will return to the original condition (zero) when the objects are at rest in their original positions. If either the mass or the spring were attached to a machine to harness the return-motion, that stored potential energy could be used to do useful tasks.

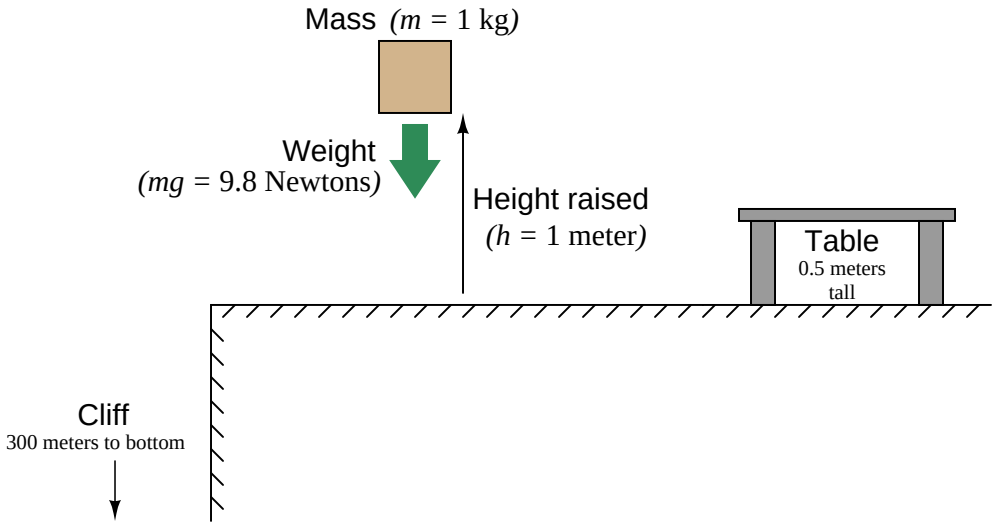
Potential energy may be similarly defined and quantified for *any* situation where we exert a force over a parallel distance, regardless of where that force or the motivating distance comes from. For instance, the static cling you experience when you pull a wool sock out of a clothes dryer is an example of a force. By pulling that sock away from another article of clothing against the force of "static cling," you are doing *work*, and storing *potential energy* in the tension between that sock and the rest of the clothing. In a similar manner, that stored energy could be released to do useful tasks if we placed the sock in some kind of machine harnessing the sock's motion as it returns to its original position on the pile of laundry, pulled by the force of static electrical attraction.

If we make use of non-mechanical means to move electric charge from one location to another, the result is no different. Moving attracting charges apart from one another means doing *work* (a force exerted over a parallel distance) and storing potential energy in that physical tension. When we use chemical reactions to move electrons from one metal plate to another in a solution, or when we spin a generator and electro-magnetically motivate electrons to seek other locations, we impart potential energy to those electrons. We could express this potential energy in the same unit as we do for mechanical systems (the *Joule*). However, it is actually more useful to express the potential energy in an electric system in terms of how many joules are available per a specific quantity of electric charge (a certain number of electrons). This measure of *specific* potential energy is simply called *electric potential* or *voltage*, and we measure it in units of *Volts*, in honor of the Italian physicist Alessandro Volta, inventor of the first electrochemical battery.

$$1 \text{ Volt} = \frac{1 \text{ Joule of potential energy}}{1 \text{ Coulomb of electric charge}}$$

In other words, if we forced 1 Coulomb's worth of electrons (6.24×10^{18} of them, to be exact) away from a positively-charged place, and did one Joule's worth of work in the process, we would have generated one Volt of electric potential.

Electric potential (voltage) and potential energy share a common, yet confusing property: both quantities are fundamentally *relative* between two physical locations. There is really no such thing as specifying a quantity of potential energy at a single location. The amount of potential energy in any system is always relative between two different points. If I lift a mass off the ground, I can specify its potential energy, *but only in relation to its former position on the ground.* The amount of energy that mass is potentially capable of releasing by free-fall depends on how far it could possibly fall. To illustrate, we will perform some "thought experiments" where we lift a 1 kilogram mass 1 meter off the ground and then drop that mass to assess its energy. That 1-kilo mass weighs 9.81 Newtons on Earth, and the distance lifted was 1 meter, so the potential energy stored in the mass is 9.81 joules, right? Consider the following scenario:



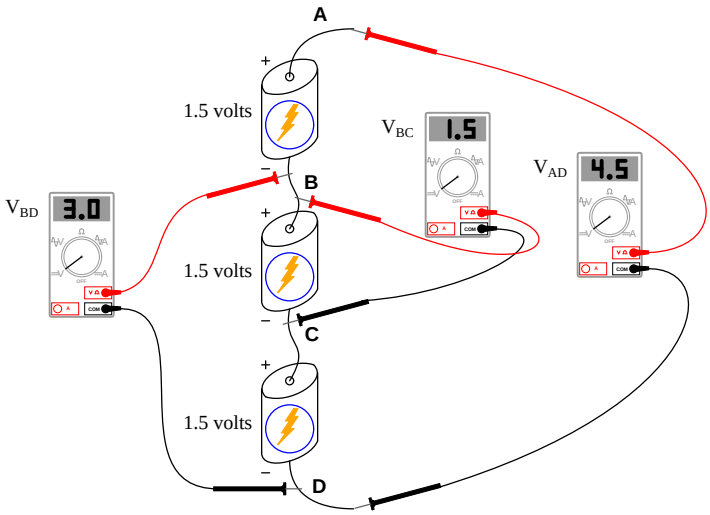
If we drop the mass over the spot we first lifted it from, it will release all the potential energy we invested in it: 9.81 joules. From this example it seems very clear that the mass possessed 9.81 joules of potential energy when lifted. However, the situation becomes more complicated if we carry the mass over to the table and release it there. Since the mass can now only fall *half* a meter, it will only release 4.9 joules of energy in the process. How much potential energy did the mass have while suspended above that table? If you argue that the mass possessed a full 9.81 joules of potential energy at that original height, then why can't we realize that full potential when dropped over the table? Can we really say that an object possess potential energy if we cannot realize that potential?

Next, imagine lifting the 1 kg mass 1 meter off the ground, and then carrying it over the edge of a cliff with a 300 meter drop. If we were to release the mass here, it would fall a full 301 meters before hitting the ground. This represents *2.95 kilojoules (kJ)* of energy! How can we possibly argue that the mass had a potential energy of 9.81 joules when lifted one meter off the ground, and then without imparting any more energy to that mass we suddenly are able to realize over 300 times as much energy?

The answer to this paradox is that potential energy is always *relative* between the object's initial height and its height after falling. It is technically incorrect to say that the mass possesses 9.81 joules of potential energy when lifted 1 meter off the ground, as though potential energy were an absolute

quantity. Instead, it is more accurate to say that the mass *gained* 9.81 joules of potential energy after being lifted 1 meter off the ground, implying that the mass could have possessed potential energy (with reference to the cliff's bottom) even while sitting on the ground (above the cliff).

Likewise, electrical voltage (as an expression of potential energy) is a quantity relative between two points. Electrical voltage is symbolized by the variable V and measured in the unit of the *Volt*. Just as we must know the mass's position relative to its falling point before we can quantify its potential energy, we must know an electric charge's position relative to its return point before we can quantify its voltage. Consider a series of batteries[1] connected as shown:



The voltage as measured between any two points directly across a single battery will be 1.5 volts:

$V_{AB} = 1.5$ volts
$V_{BC} = 1.5$ volts
$V_{CD} = 1.5$ volts

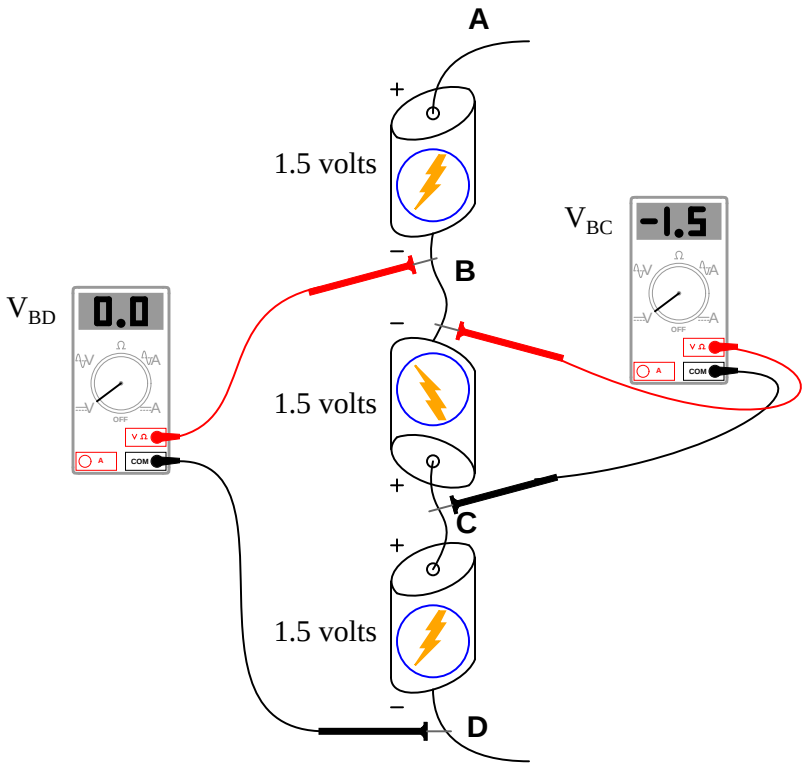
If, however, we span more than one battery with our voltmeter connections, our voltmeter will register more than 1.5 volts:

$V_{AC} = 3.0$ volts
$V_{BD} = 3.0$ volts
$V_{AD} = 4.5$ volts

Thus we see there is no such thing as "voltage" at a single point in a circuit. The concept of voltage has meaning only *between* pairs of points in a circuit, just as the concept of potential energy for a mass has meaning only *between* two physical locations: where the mass is, and where it may potentially fall.

[1] A *battery* is an electrochemical device producing an electrical voltage as the result of a chemical reaction.

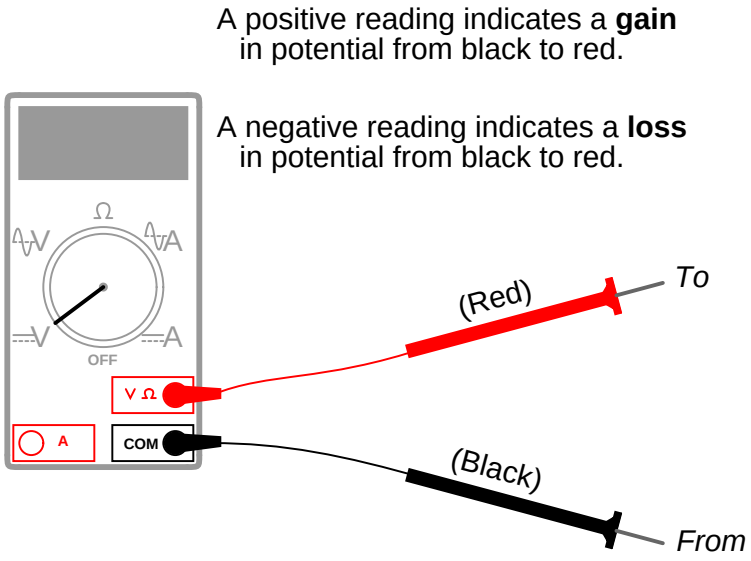
Things get interesting when we connect voltage sources in different configurations. Consider the following example, identical to the previous illustration except the middle battery has been reversed:



Note the "+" and "−" signs next to the ends of the batteries. These signs show the *polarity* of each battery's voltage. Also note how the two voltmeter readings are different from before. Here we see an example of *negative potential* with the middle battery connected in opposition to the other two batteries. While the top and bottom batteries are both "lifting" electric charges to greater potential (going from point **D** to point **A**), the middle battery is decreasing potential from point **C** to point **B**. It's like taking a step forward, then a step back, then another step forward. Or, perhaps more appropriately, like lifting a mass 1.5 meters up, then setting it down 1.5 meters, then lifting it 1.5 meters up again. The first and last steps accumulate potential energy, while the middle step releases potential energy.

This explains why it is important to install multiple batteries the same way into battery-powered devices such as radios and flashlights. The batteries' voltages are supposed to add to a make a larger total required by the device. If one or more batteries are placed backwards, potential will be lost instead of gained, and the device will not receive enough voltage.

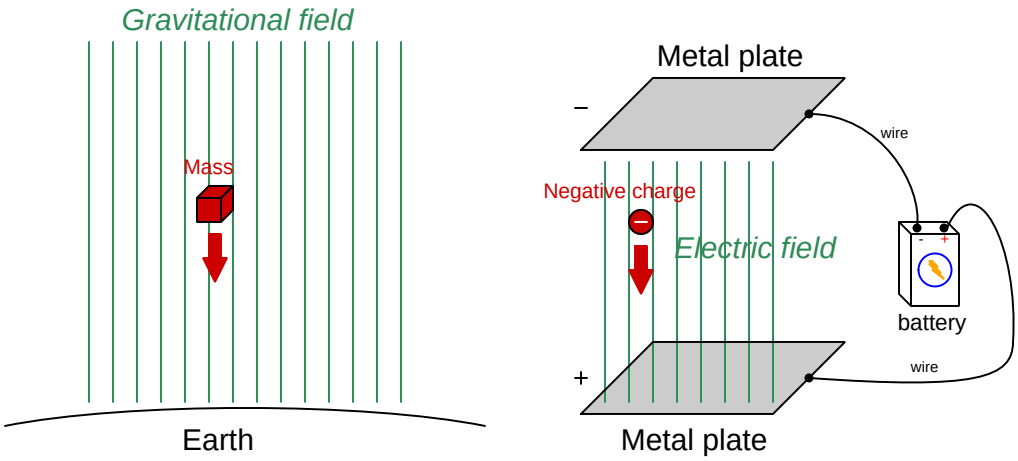
Here we must pay special attention to how we use our voltmeter, since polarity matters. All voltmeters are standardized with two colors for the test leads: red and black. To make sense of the voltmeter's indication, especially the positive or negative *sign* of the indication, we must understand what the red and black test lead colors mean:



Connecting these test leads to different points in a circuit will tell you whether there is potential gain or potential loss from one point (black) to the other point (red).

4.2 Electrical current

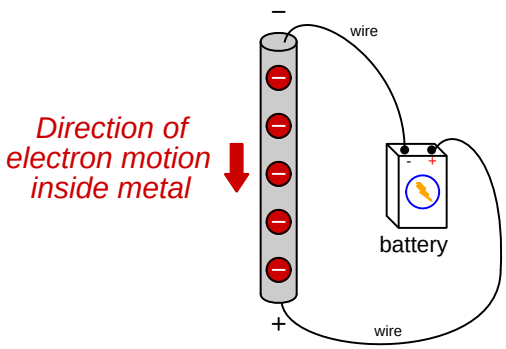
Current is the name we give to the motion of electric charges from a point of high potential to a point of low potential. All we need to form an electric current is a source of potential (voltage) and some electric charges that are free to move between the poles of that potential. For instance, if we connected a battery to two metal plates, we would create an electric field between those plates, analogous to a gravitational field except it only acts on electrically charged objects, while gravity acts on anything with mass. A free charge placed between those plates would "fall" toward one of the plates just as a mass would fall toward a larger mass:



An electric charge will "fall" in an electric field just as a mass will fall in a gravitational field.

Some solid substances, most notably metals, have very mobile electrons. That is, the outer (valence) electrons are very easily dislodged from the parent atoms to drift to and fro throughout the material. In fact, the electrons within metals are so free that physicists sometimes refer to the structure of a metal as atomic nuclei "floating in a sea of electrons". The electrons are almost fluid in their mobility throughout a solid metal object, and this property of metals may be exploited to form definite pathways for electric currents. This is why electrical wires are made of metal such as copper or aluminum: substances with a high degree of electron mobility. Any substance whose electrons are mobile is called a *electrical conductor*, while any substance lacking mobile electrons is called an *electrical insulator*.

If the poles of a voltage source are joined by a conductor, the free electrons within that conductor will drift toward the positive pole (electrons having a negative charge, opposite charges attracting one another). For each electron reaching the positive pole, an electron exits the negative pole of the source to replenish the total number of electrons in the flow:

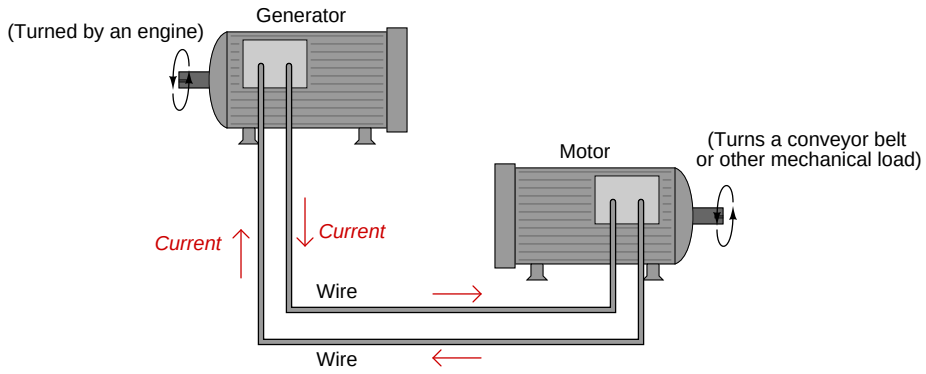


If the source of this voltage is continually replenished by chemical energy, mechanical energy, or some other form of energy, the free electrons will continually loop around this circular path. We call this unbroken path an *electric circuit*. The drifting motion of electrons in a circuit has the same average rate of flow (current) at all points in that circuit, because there is only one pathway for the current. You may think of this like liquid flowing through a circular loop of pipe: since there is only one pathway for the liquid to flow, the rate of flow at all points in that pathway must be the same.

We typically measure the amount of current in a circuit by the unit of *Amperes*, or *Amps* for short (named in honor of the French physicist André Ampère, and symbolize it in formulae using the variable I. One ampere of current is equal to one coulomb of electric charge (6.24×10^{18} electrons) moving past a point in a circuit for every second of time.

Like masses falling toward a source of gravity, these electrons continually "fall" toward the positive pole of a voltage source. After arriving at that source, the energy imparted by that source "lifts" the electrons to a higher potential state where they once again "fall down" to the positive pole through the circuit.

Like rising and falling masses in a gravitational field, these electrons act as carriers of energy within the electric field of the circuit. This is very useful, as we can use them to convey energy from one place to another, using metal wires as conduits for this energy. This is the basic idea behind electric power systems: a source of power (a *generator*) is turned by some mechanical engine (windmill, water turbine, steam engine, etc.), creating an electric potential. This potential is then used to motivate free electrons inside the metal wires to drift in a common direction. The electron drift is conveyed in a circuit through long wires, where they can do useful work at a *load* device such as an electric motor, light bulb, or heater.



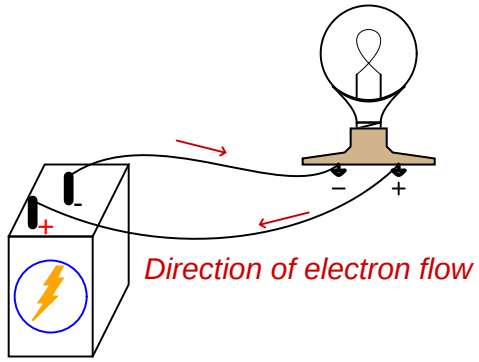
Given the proper metal alloys, the friction that electrons experience within the metal wires may be made very small, allowing nearly all the energy to be expended at the load (motor), with very little wasted along the path (wires). This makes electricity the most efficient means of energy transport known.

The electric currents common in electric power lines may range from hundreds to thousands of amperes. The currents conveyed through power receptacles in your home typically are no greater than 20 amperes. The currents in the small battery-powered circuits you will build are even less: fractions of an ampere. For this reason, we commonly use the metric prefix *milli* (one one-thousandth) to express these small currents. For instance, 10 milliamperes is 0.010 amperes, and 500 milliamperes is one-half of an ampere.

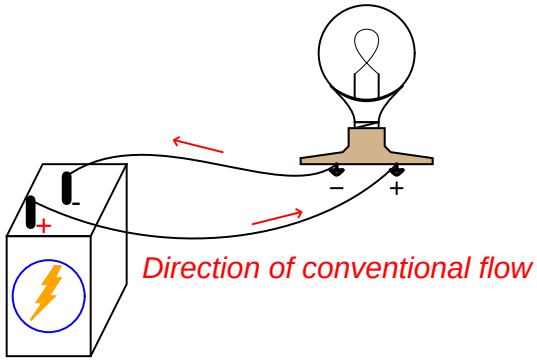
4.2.1 Electron versus conventional flow

When Benjamin Franklin proposed his single-fluid theory of electricity, he defined "positive" and "negative" as the surplus and deficiency of electric charge, respectively. These labels were largely arbitrary, as Mr. Franklin had no means of identifying the actual nature of electric charge carriers with the primitive test equipment and laboratory techniques of his day. As (bad) luck would have it, his hypothesis was precisely opposite of the truth for metallic conductors, where electrons are the dominant charge carrier.

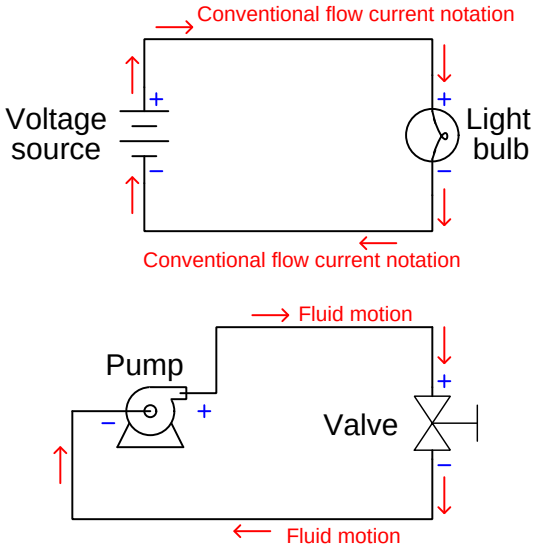
This means that in an electric circuit consisting of a battery and a light bulb, electrons slowly move from the negative side of the battery, through the metal wires, through the light bulb, and on to the positive side of the battery as such:



Unfortunately, scientists and engineers had grown accustomed to Franklin's false hypothesis long before the true nature of electric current in metallic conductors was discovered. Their preferred notation was to show electric current flowing from the positive pole of a source, through the load, returning to the negative pole of the source:

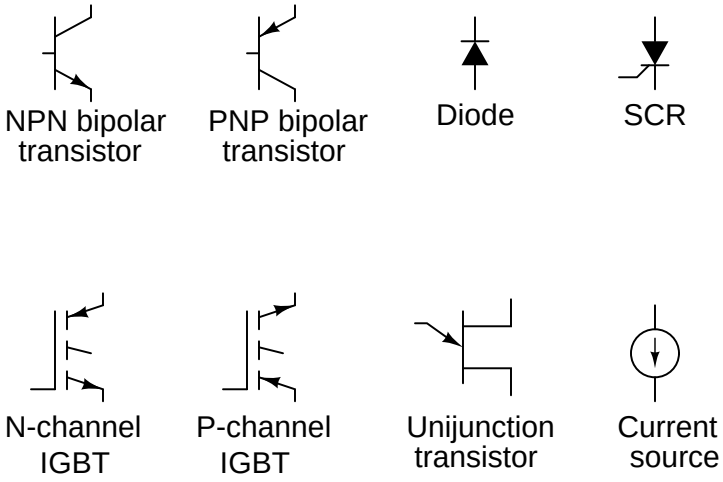


This relationship between voltage polarity marks and conventional flow current tends to make more intuitive sense than electron flow notation, because it is reminiscent of fluid pressure and flow direction:



If we take the "+" sign to represent *more* pressure and the "−" sign to represent *less* pressure, it makes perfect sense that fluid should move from the high-pressure (discharge) port of the pump through the hydraulic "circuit" and back to the low-pressure (suction) port of the pump. It also makes perfect sense that the upstream side of the valve (a fluid restriction) will have a greater pressure than the downstream side of the valve. In other words, conventional flow notation best honors Mr. Franklin's original intent of modeling current as though it were a fluid, even though he was later proven to be mistaken in the case of metallic conductors where electrons are the dominant charge carrier.

This convention was so well-established in the electrical engineering realm that it held sway despite the discovery of electrons. Engineers, who create the symbols used to represent the electronic devices they invent, consistently chose to draw arrows in the direction of conventional flow rather than electron flow. In each of the following symbols, the arrow heads point in the direction that *positive* charge carriers would move (opposite the direction that electrons actually move):



This stands in contrast to electronics technicians, who historically have been taught using electron flow notation. I remember sitting in a technical school classroom being told by my teacher to always imagine the electrons moving *against the arrows* of the devices, and wondering why it mattered.

It is truly a sad situation when the members of two branches within the same field do not agree on something as fundamental as the convention used to denote flow in diagrams. It is even worse when people within the field argue over which convention is best. So long as one is consistent with their convention and with their thinking, *it does not matter!* Many fine technologists may be found on either side of this "fence," and some are adept enough to switch between both without getting confused.

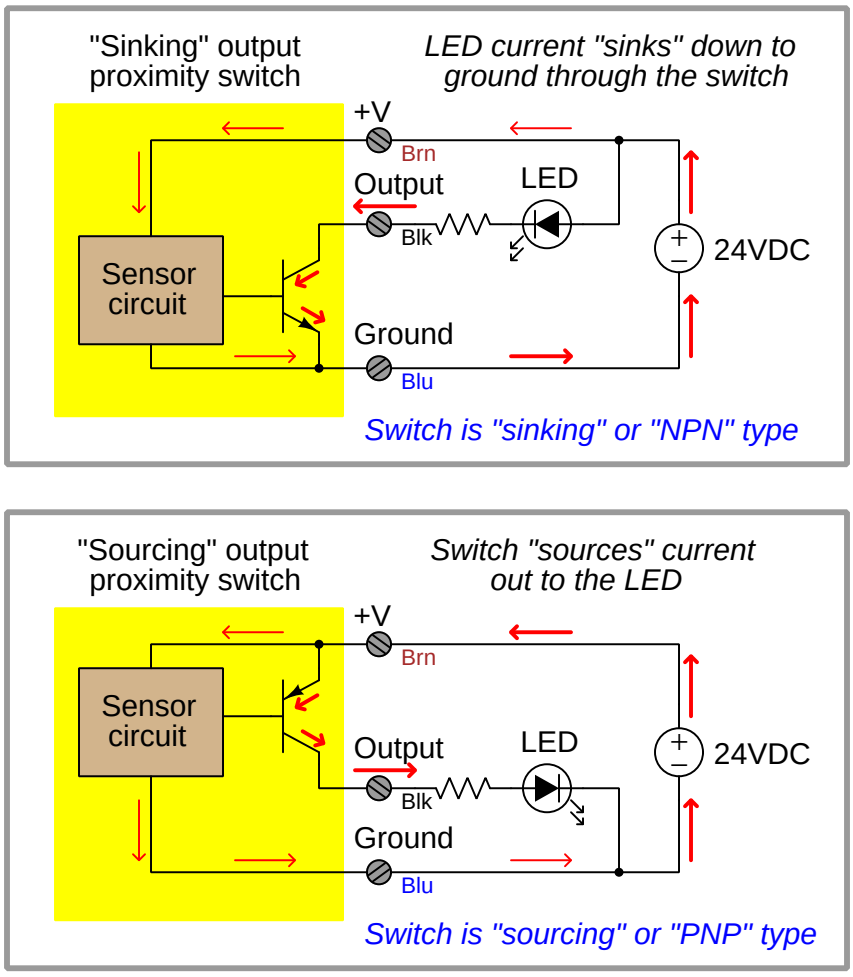
My personal preference is conventional flow notation. The only objective arguments I have in favor of this preference are as follows:

- Conventional flow notation is a closer analogue to fluid flow in pneumatic, hydraulic, and process flow systems. Since instrument technicians need to understand the relationships between pressure and flow in fluid systems as well as electrical circuits, using conventional flow notation for electrical circuits helps to reinforce these analogous concepts.
- Conventional flow notation is the *standard* for modern manufacturers' documentation (reference manuals, troubleshooting guides, datasheets, etc.)[2].
- Conventional flow notation is consistent with the "right-hand rule" for vector cross products (which are essential for understanding electromagnetics at advanced academic levels). The so-called "left-hand rule" taught to students learning electron flow notation is mathematically wrong, and must be un-learned if the student ever progresses to the engineering level in his or her studies.
- Conventional flow notation matches all device arrows; no need to "go against the arrow" when tracing current in a schematic diagram.
- Conventional flow notation makes sense of the descriptive terms *sourcing* and *sinking*.

This last point merits further investigation. The terms "sourcing" and "sinking" are often used in the study of digital electronics to describe the direction of current in a switching circuit. A circuit that "sources" current to a load is one where the direction of conventional flow points outward from the sourcing circuit to the load device. These terms are also used to characterize the DC input and output modules of certain industrial control equipment such as programmable logic controllers (PLCs), and so any technologist working with such equipment will need to properly identify and connect these module types based on the directions of electric current for which they are designed.

[2] I have yet to read a document of any kind written by an equipment manufacturer using electron flow notation, and this is after scrutinizing literally hundreds of documents looking for this exact detail! For the record, though, most technical documents do not bother to draw a direction for current at all, leaving it to the imagination of the reader instead. It is only when a direction must be drawn that one sees a strong preference in industry for conventional flow notation.

For example, here are two schematic diagrams showing two different kinds of electronic proximity switch. The first switch *sinks* current in from the LED through its output terminal, through its transistor, and down to ground. The second switch *sources* current from the positive supply terminal through its transistor and out to the LED through its output terminal (note the direction of the thick arrow near the output screw terminal in each circuit):



These terms simply make no sense when viewed from the perspective of electron flow notation. If you were to actually trace the directions of the electrons, you would find that a device "sourcing" current has electrons flowing *into* its connection terminal, while a device "sinking" current sends electrons *out* to another device where they travel (up) to a point of more positive potential.

In fact, the association between conventional flow notation and sourcing/sinking descriptions is so firm that I have yet to see a professionally published textbook on digital circuits that uses electron flow[3]. This is true even for textbooks written for technicians and not engineers!

[3]If by chance I have missed anyone's digital electronics textbook that does use electron flow, please accept my

Once again, though, it should be understood that either convention of current notation is adequate for circuit analysis. I dearly wish this horrible state of affairs would come to an end, but the plain fact is it will not. Electron flow notation may have the advantage of greater correspondence to the actual state of affairs (in the vast majority of circuits), but conventional flow has the weight of over a hundred years of precedent, cultural inertia, and convenience. No matter which way you choose to think, at some point you will be faced with the opposing view.

Choose the notation you like best, and may you live long and prosper.

4.3 Electrical sources and loads

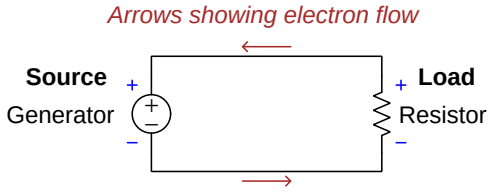
By definition, a *source* is a device delivering energy into a system, while a *load* is a device extracting energy from a system. Examples of typical electrical sources include generators, photovoltaic cells, thermopiles, and primary-cell batteries. These devices create electrical voltage, which in turn motivates electrical current to flow in a circuit. Examples of typical electrical loads include resistors, lamps, and electric motors. These devices resist the flow of electrical current through them, creating a voltage *drop* as a result.

In a working circuit, electrical sources and loads may be easily distinguished by comparison of their current directions and voltage polarities. An electrical source always manifests a voltage polarity in a direction *aiding* the direction of charge flow; i.e. a source is "pushing" the current along. An electrical load always manifests a voltage polarity in a direction *opposing* the direction of charge flow; i.e. a load "resists" the current.

The way in which we designate the direction of current (charge flow) becomes very important here. Since there are two commonly accepted notations – electron flow and "conventional" flow, exactly opposite of each other – it is easy to become confused.

apologies. I can only speak of what I have seen myself.

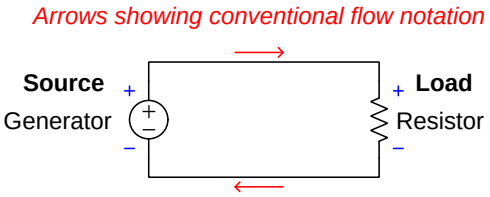
First we see a diagram showing a source and a load, using electron flow notation. Electrons, being negatively charged particles, are repelled by the negative (−) poles of both source and load, and attracted to the positive (+) poles of both source and load. The difference between source and load is that the source device *motivates* the flow of electrons while the load device *resists* the flow of electrons:



Electrons are repelled by the (-) poles
and attracted to the (+) poles

In the case of the source (battery), the polarity of the voltage works *for* the direction of charge motion. In the case of the load (resistor), the polarity of the voltage drop works *against* the direction of charge motion.

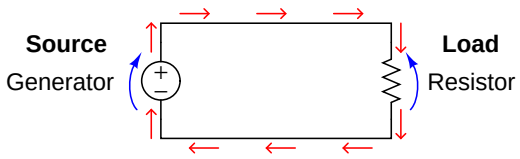
Next we see a diagram showing the same source and load, this time using "conventional" flow notation to designate the direction of current. Here we must imagine positively-charged carriers moving through the wires instead of electrons. These positive charge carriers are repelled by any positive (+) pole and attracted to any negative (−) pole. Viewed in this light, we see the exact same principle at work: the source device is seen to *motivate* the flow of these positive charge carriers while the load device *resists* the flow:



Positive charge carriers are repelled by the
(+) poles and attracted to the (-) poles

Despite using a different notation for charge motion, the concept of source and load remains the same. In the case of the source (battery), the polarity of the voltage works *for* the direction of charge motion. In the case of the load (resistor), the polarity of the voltage drop works *against* the direction of charge motion.

An alternative notation for voltage (other than using "+" and "−" symbols) that many students find particularly illustrative is the use of curved arrows, where the tip of the curved arrow is the positive pole and the tail of the curved arrow is the negative pole. This notation is intended to be used when the direction of current (using "straight" or "angular" arrows) is shown using conventional flow notation:



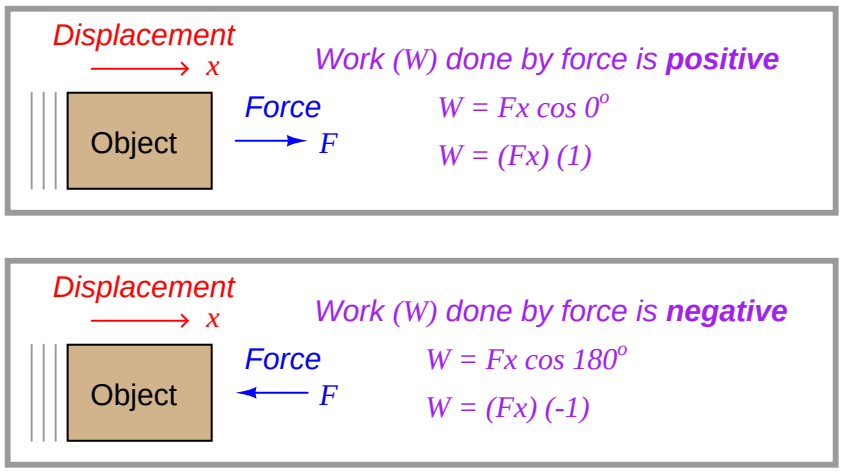
Curved arrows show direction of voltage
(electro-motive force, or EMF)

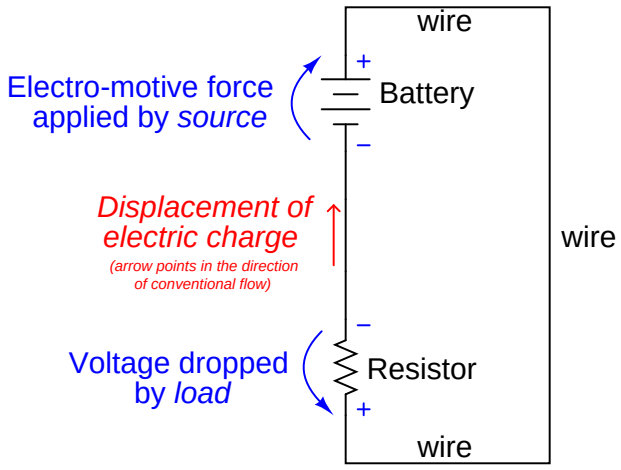
Straight arrows show direction of current
(conventional flow assumed; positive charges moving)

Using arrows to represent voltage polarity in addition to using arrows to represent current direction is highly intuitive. It shows which way each component in the DC circuit is "pushing" in relation to the flow of charge carriers. Note how the source's voltage arrow points in the same direction as the current: this means the source is motivating the current, causing charge carriers to flow in this circuit. Note how the resistor's voltage arrow points opposite to the direction of current: this means the resistor is opposing the current, in a sense "fighting against" the flow of charge carriers. This comparison of voltage-arrow versus current-arrow direction makes the distinction between sources and loads rather obvious: sources push in the direction of current while loads push against the direction of current.

I personally lament the obscurity of this "curved-arrow" notation for voltage, as it greatly aids comprehension of this critically important distinction between sources and loads. When the voltage and current arrows point in the same direction, it means the component in question is motivating charge carriers along and therefore imparts energy to the circuit. When voltage and current arrows point in opposite directions, it means the component in question opposes charge carrier motion and therefore acts to extract energy from the circuit.

This directly relates to the fundamental physics concept of *work*, specifically in relation to the mathematical sign of work being a function of the relative angle between force and displacement. When a force acts in the same direction as motion, the work done is positive; when a force acts in the opposite direction as motion, the work done is negative:



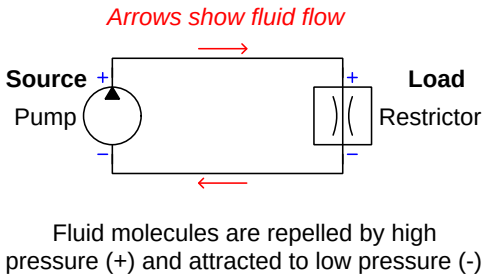


$W_{battery}$ = positive value, because its force is in the same direction as the current

$W_{resistor}$ = negative value, because its force opposes the direction of the current

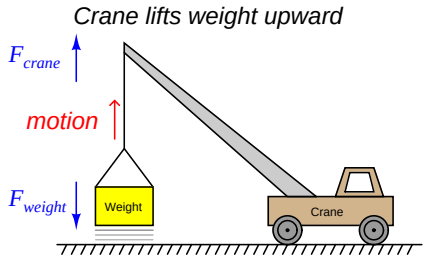
Positive work represents an infusion of energy into a system (source), while negative work represents an extraction of energy from that system (load).

If we examine a hydraulic system, where a pump pushes fluid around a pipe loop and an orifice (called a "restrictor") restricts the flow of this fluid, we see this same concept in action: the pump's pressures at its discharge and suction ports work *for* the direction of fluid flow, while the pressures at the upstream and downstream ports of the restrictor work *against* the direction of fluid flow. The pump acts as a power *source* in this hydraulic "circuit" (infusing energy into the system) while the restrictor acts as a power *load* (extracting energy from the system):



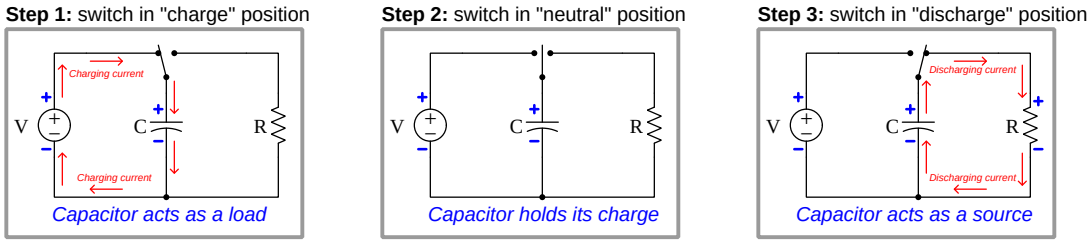
Fluid molecules are repelled by high pressure (+) and attracted to low pressure (-)

We may even see this concept revealed in a simple mechanical system where work is being done. Examine the case of a crane lifting a heavy weight into the air, shown below. As the crane lifts the weight upward, the crane's upward force on the weight is clearly working *for* the direction of motion, while the weight's downward force against the crane is clearly working *against* the direction of motion:



Thus, the crane is doing positive work (acting as a *source*, infusing potential energy into the weight) while the weight is doing negative work (acting as a *load*, absorbing potential energy from the crane). A mathematically rigorous way to demonstrate this is to calculate the work done by each using the formula $W = \vec{F} \cdot \vec{x}$ or $W = Fx\cos\theta$. Since the crane's force and motion vectors both point in the same direction, $\theta = 0$ and work is a positive quantity Fx. The weight's force vector, however, points 180° away from the motion vector, and so its work calculation is $Fx\cos(180^o)$ or $-Fx$.

Some electric components have the ability to act as sources and loads at different times. Both capacitors (see section 4.12 starting on page 346) and inductors (see section 4.13 starting on page 350) have the ability to temporarily contribute to and extract energy from electrical circuits, both having the ability to act as energy *storage* devices. One of the key concepts necessary to grasp the energy-storing abilities of capacitors and inductors is being able to recognize sources and loads at a glance based on the relationship between voltage polarity and charge motion. A set of three schematic diagrams shows how a capacitor is able to play the role of either source or load depending on what other component it is connected to:



Rechargeable batteries (called "secondary-cell" batteries as opposed to "primary-cell" batteries which cannot be recharged) may also behave as either sources or loads depending on external conditions. If a secondary-cell battery is connected to a resistor, the battery will discharge its energy (i.e. act as a source) while the resistor will dissipate that energy (i.e. act as a load). If a depleted secondary-cell battery is connected to an electrical generator of greater voltage, the generator will source energy to the battery while the re-charging battery will load down the generator (i.e. conventional flow entering the battery's positive terminal and exiting the negative terminal).

Another practical benefit of clearly comprehending the distinction between electrical sources and electrical loads is being able to understand and troubleshoot 4-20 mA signal "loop" circuits used extensively in industrial instrumentation, especially circuits containing 2-wire ("loop-powered") process transmitters. A "2-wire transmitter" is a device designed to regulate the amount of electrical current through it to a value determined by some physical variable such as a sensed pressure, temperature, or flow rate. The purpose of such a device is to represent that physical measurement in the form of an electric current that may be carried far distances through wires. What makes this device so troublesome for people to understand is that despite its function to *set* the value of current in the circuit, it is actually an electrical *load* and not an electrical *source* as one might assume. That is, a 2-wire transmitter relies wholly on some other source of electrical power in the circuit to function, although the transmitter solely defines how much current will flow in the circuit by virtue of its function as a regulator. For more information on this subject, refer to section 13.5 starting on page 893.

4.4 Electrical power

Since we often use electricity to convey energy from one location to another, it is important to be able to quantify and calculate the rate at which energy is delivered by an electric circuit. The *rate* at which energy transfers from one location to another is technically referred to as *power* (P), and it is typically[4] measured in the unit of the *watt* (1 watt = 1 joule of energy per second of time).

The total amount of power (energy per unit time) carried by an electric current is proportional to the strength of that current as well as the amount of potential (voltage) between the poles of the electrical source. Thus, power is equal to the product of current and voltage in a formula called *Joules' Law*:

$$P = IV$$

Where,
P = Power in watts (joules of energy per second of time)
I = Current in amperes
V = Voltage in volts

We may prove the correctness of this formula by verifying all the units of measurement agree:

$$P = IV$$

$$[\text{Watts}] = [\text{Amperes}] \times [\text{Volts}] \qquad \text{or} \qquad [\text{W}] = [\text{A}][\text{V}]$$

$$\left[\frac{\text{Joules}}{\text{Seconds}}\right] = \left[\frac{\text{Coulombs}}{\text{Seconds}}\right] \times \left[\frac{\text{Joules}}{\text{Coulombs}}\right] \qquad \text{or} \qquad \left[\frac{\text{J}}{\text{s}}\right] = \left[\frac{\text{C}}{\text{s}}\right]\left[\frac{\text{J}}{\text{C}}\right]$$

Note how the basic units for power (joules per second) are indeed equal to the product of voltage (joules per coulomb) and current (coulombs per second). This process of checking for agreement between units of measurement in a physics formula is called *dimensional analysis*.

[4] Although the unit of the "watt" is commonly used for electrical power, other units are valid as well. The British unit of *horsepower* is every bit as valid for expressing electrical power as "watts," although this usage is less common. Likewise, the "watt" may be used to express measurements of non-electrical power as well, such as the mechanical power output of an engine. European automobile manufacturers, for example, rate the power output of their cars' engines in *kilowatts*, as opposed to American automobile manufacturers who rate their engines in *horsepower*. This choice of units is strictly a cultural convention, since any valid unit for power may be applied to any form of energy rate.

4.5 Electrical resistance and Ohm's Law

To review, *voltage* is the measure of potential energy available to electric charges. *Current* is the uniform drifting of electric charges in response to a voltage. We can have a voltage without having a current, but we cannot have a current without first having a voltage to motivate it[5]. Current without voltage would be equivalent to motion without a motivating force.

When electric charges move through a material such as metal, they will naturally encounter some friction, just as fluid moving through a pipe will inevitably encounter friction[6]. We have a name for this friction to electrical charge motion: *resistance*. Like voltage and current, resistance has its own special unit of measurement: the *ohm*, named in honor of the German physicist Georg Simon Ohm.

At this point it would be good to summarize and compare the symbols and units we use for voltage, current, and resistance:

Quantity	**Algebraic symbol**	**Unit**	**Unit abbreviation**
Voltage	V (or E)	Volt	V
Current	I	Ampere (or Amp)	A
Resistance	R	Ohm	Ω

Ohm defined resistance as the mathematical ratio between applied voltage and resulting current. This formula become known as *Ohm's Law*, perhaps the most basic formula in all of electrical science (shown here in three different forms, each one solving for a different variable):

$$R = \frac{V}{I} \qquad V = IR \qquad I = \frac{V}{R}$$

Verbally expressed, resistance is how much voltage it takes to force a certain rate of current through a conductive material. Many materials have relatively stable resistances, while others do not. Devices called *resistors* are sold which are manufactured to possess a very precise amount of resistance, for the purpose of limiting current in circuits (among other things).

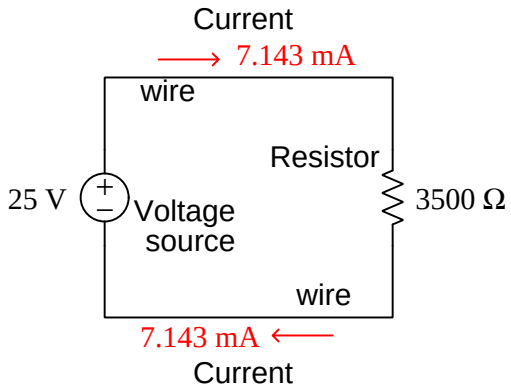
Here is an example of Ohm's Law in action: calculate the amount of current in a circuit with a voltage source of 25 V and a total resistance of 3500 Ω. Taking 25 volts and dividing by 3500 ohms, you should arrive at a result of 0.007143 amperes, or 7.143 milliamperes (7.143 mA).

One of the most challenging aspect of Ohm's Law is remembering to *keep all variables in context*. This is a common problem for many students when studying physics as well: none of the equations learned in a physics class will yield the correct results unless all the variables relate to the same object or situation. For instance, it would make no sense to try to calculate the kinetic energy of a moving object ($E = \frac{1}{2}mv^2$) by taking the mass of one object (m) and multiplying it by the square of the velocity of some *other* object (v^2). Likewise, with Ohm's Law, we must make sure the voltage, current, and resistance values we are using all relate to the same portion of the same circuit.

[5] Except in the noteworthy case of *superconductivity*, a phenomenon occurring at extremely low temperatures.

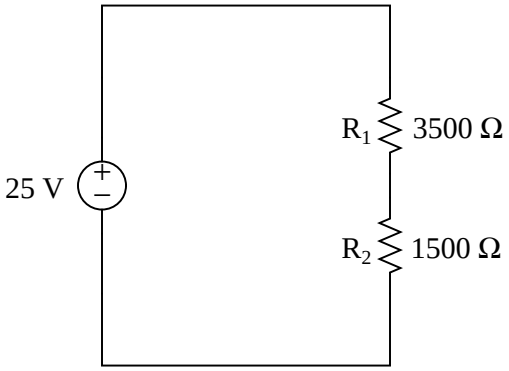
[6] Except in the noteworthy case of *superfluidity*, another phenomenon occurring at extremely low temperatures.

If the circuit in question has only one source of voltage, one resistance, and one path for current, we cannot mis-apply Ohm's Law. Expressing the previous example in a schematic diagram:

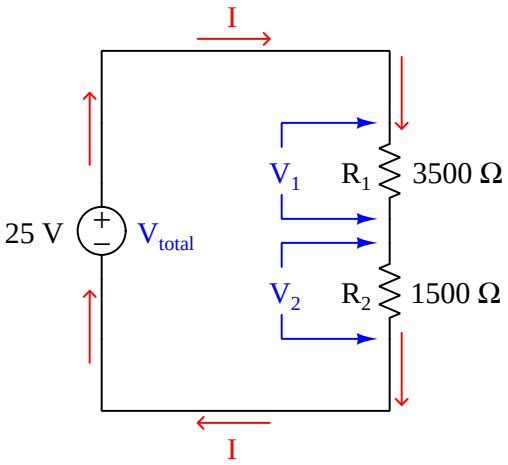


Note: arrows point in the direction of conventional flow

However, if we look at a more complex circuit, we encounter the potential for mis-application of Ohm's Law because we have multiple resistances in the circuit rather than just one resistance:



Which resistance do we use to calculate current in this circuit? Do we divide our 25 volts by 3500 ohms like we did last time, or do we divide it by 1500 ohms, or something entirely different? The answer to this question lies in the identification of voltages and currents. We know that the 25 volt potential will be impressed across the *total* of the two resistances R_1 and R_2, and since there is only one path for current they must share the same current. Thus, we actually have *three* voltages (V_1, V_2, and V_{total}), *three* resistances (R_1, R_2, and R_{total}), and only *one* current (I):



Note: arrows point in the direction of conventional flow

Using the $V = IR$ form of Ohm's Law to relate these three voltages (V_1, V_2, and V_{total}) to the one current (I), we end up with three equations for this circuit:

$$V_1 = IR_1$$

$$V_2 = IR_2$$

$$V_{total} = IR_{total} = I(R_1 + R_2)$$

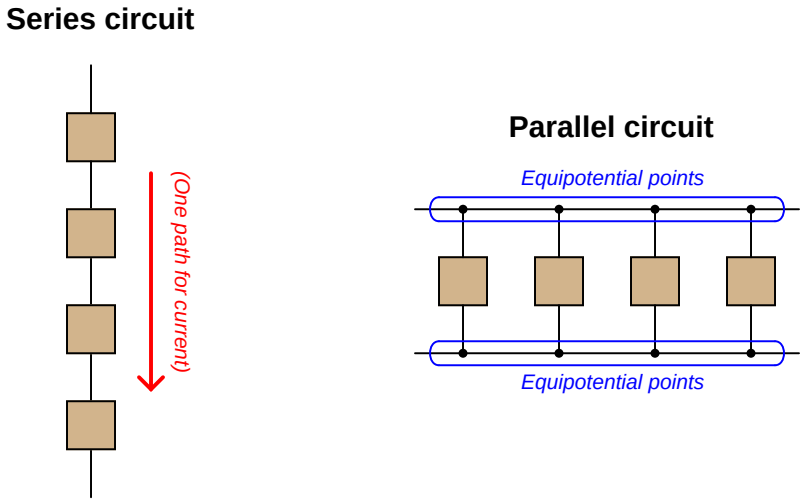
We can only solve for one unknown variable at a time in any equation. This means we cannot solve for V_1 yet, because although we do know the value of resistor R_1 (3500 ohms), we do not yet know the circuit's current (I). Likewise for V_2, because we do not yet know the value of I. However, the third equation is solvable, since we happen to know the total voltage as well as both resistor values, leaving circuit current I as the only unknown variable. Manipulating this equation and solving for I:

$$I = \frac{V_{total}}{R_{total}} = \frac{V_{total}}{R_1 + R_2} = \frac{25 \text{ V}}{3500\ \Omega + 1500\ \Omega} = 0.005 \text{ A} = 5 \text{ mA}$$

Now that we know the amount of current in this circuit, we may solve for V_1 and V_2 in the other equations to find that the voltage across resistor R_1 is 17.5 volts, and that the voltage across resistor R_2 is 7.5 volts.

4.6 Series versus parallel circuits

In addition to Ohm's Law, we have a set of rules describing how voltages, currents, and resistances relate in circuits comprised of multiple resistors. These rules fall neatly into two categories: *series* circuits and *parallel* circuits. The two circuit types are shown here, with squares representing any type of two-terminal electrical component:

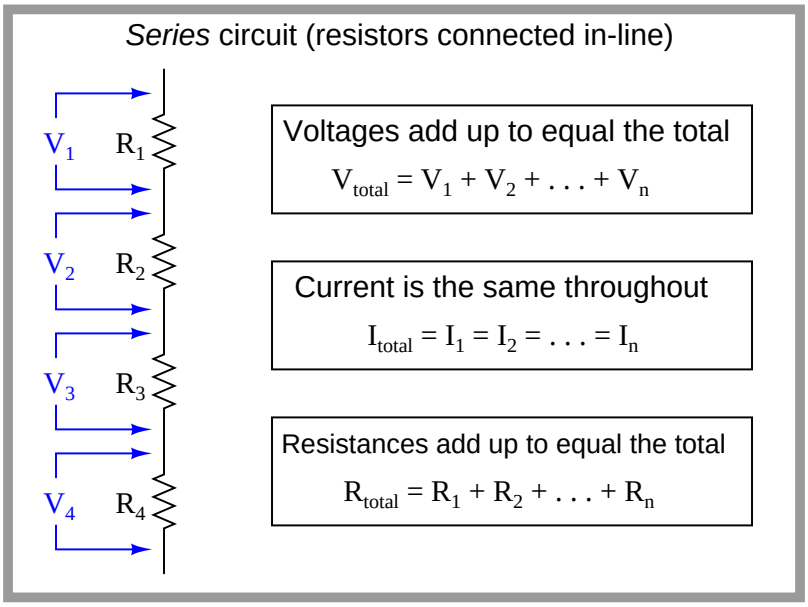


The defining characteristic of a series electrical circuit is it provides just one path for current. This means there can be only one value for current anywhere in the circuit, the exact same current for all components at any given time[7]. The principle of current being the same everywhere in a series circuit is actually an expression of a more fundamental law of physics: the *Conservation of Charge*, which states that electric charge cannot be created or destroyed. In order for current to have different values at different points in a series circuit indefinitely, electric charge would have to somehow appear and disappear to account for greater rates of charge flow in some areas than in others. It would be the equivalent of having different rates of water flow at different locations along one length of pipe[8].

[7] Interesting exceptions do exist to this rule, but only on very short time scales, such as in cases where we examine the a transient (pulse) signal nanosecond by nanosecond, and/or when very high-frequency AC signals exist over comparatively long conductor lengths.

[8] Those exceptional cases mentioned earlier in the footnote are possible only because electric charge may be temporarily stored and released by a property called *capacitance*. Even then, the law of charge conservation is not violated because the stored charges re-emerge as current at later times. This is analogous to pouring water into a bucket: just because water is poured into a bucket but no water leaves the bucket does not mean that water is magically disappearing. It is merely being stored, and can re-emerge at a later time.

Series circuits are defined by having only one path for current, and this means the steady-state current in a series circuit must be the same at all points of that circuit. It also means that the sum of all voltages dropped by load devices must equal the sum total of all source voltages, and that the total resistance of the circuit will be the sum of all individual resistances:



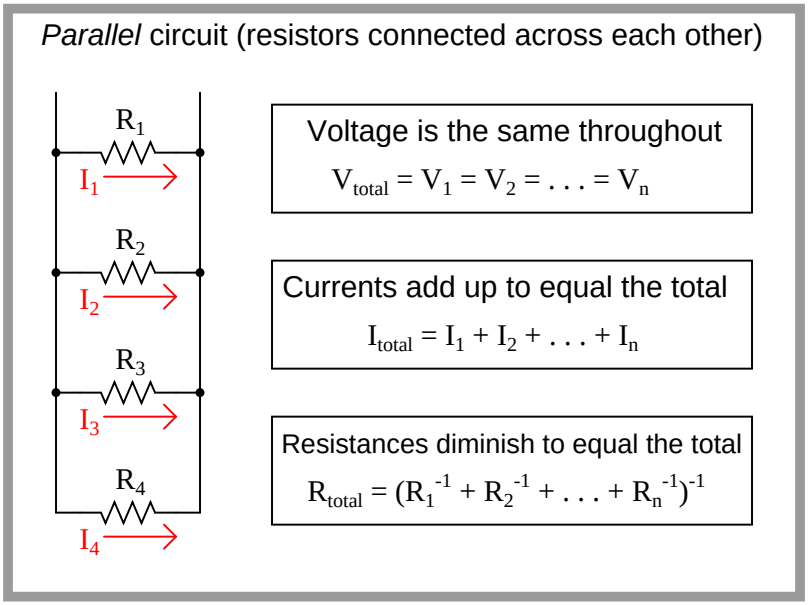
The defining characteristic of a parallel circuit, by contrast, is that all components share the same two equipotential points. "Equipotential" simply means "at the same potential" which points along an uninterrupted conductor must be[9]. This means there can be only one value of voltage anywhere in the circuit, the exact same voltage for all components at any given time[10]. The principle of voltage being the same across all parallel-connected components is (also) an expression of a more fundamental law of physics: the *Conservation of Energy*, in this case the conservation of specific potential energy which is the definition of voltage. In order for voltage to differ between parallel-connected components, the potential energy of charge carriers would have to somehow appear and disappear to account for lesser and greater voltages. It would be the equivalent of having a "high spots" and "low spots" of water mysteriously appear on the quiet surface of a lake, which we know cannot happen because water has the freedom to move, meaning any high spots would rush to fill any low spots[11].

[9] An ideal conductor has no resistance, and so there is no reason for a difference of potential to exist along a pathway where nothing stands in the way of charge motion. If ever a potential difference developed, charge carriers within the conductor would simply move to new locations and neutralize the potential.

[10] Again, interesting exceptions do exist to this rule on very short time scales, such as in cases where we examine the a transient (pulse) signal nanosecond by nanosecond, and/or when very high-frequency AC signals exist over comparatively long conductor lengths.

[11] The exceptional cases mentioned in the previous footnote exist only because the electrical property of *inductance* allows potential energy to be stored in a magnetic field, manifesting as a voltage different along the length of a conductor. Even then, the Law of Energy Conservation is not violated because the stored energy re-emerges at a later time.

The sum of all component currents must equal the total current in a parallel circuit, and total resistance will be *less* than the smallest individual resistance value:



The rule for calculating total resistance in a parallel circuit perplexes many students with its weird compound reciprocal notation. There is a more intuitive way to understand this rule, and it involves a different quantity called *conductance*, symbolized by the letter G.

Conductance is defined as the reciprocal of resistance; that is, a measure of how *easily* electrical charge carriers may move through a substance. If the electrical resistance of an object doubles, then it now has *half* the conductance it did before:

$$G = \frac{1}{R}$$

It should be intuitively apparent that conductances add in parallel circuits. That is, the total amount of conductance for a parallel circuit must be the sum total of all individual conductances, because the addition of more conductive pathways must make it easier overall for charge carriers to move through the circuit. Thus,

$$G_{total} = G_1 + G_2 + \cdots + G_n$$

The formula shown here should be familiar to you. It has the same form as the total resistance formula for series circuits. Just as resistances add in series (more series resistance makes the overall resistance to current increase), conductances add in parallel (more conductive branches makes the overall conductance increase).

Knowing that resistance is the reciprocal of conductance, we may substitute $\frac{1}{R}$ for G wherever we see it in the conductance equation:

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

Now, to solve for R_{total}, we need to reciprocate both sides:

$$R_{total} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}}$$

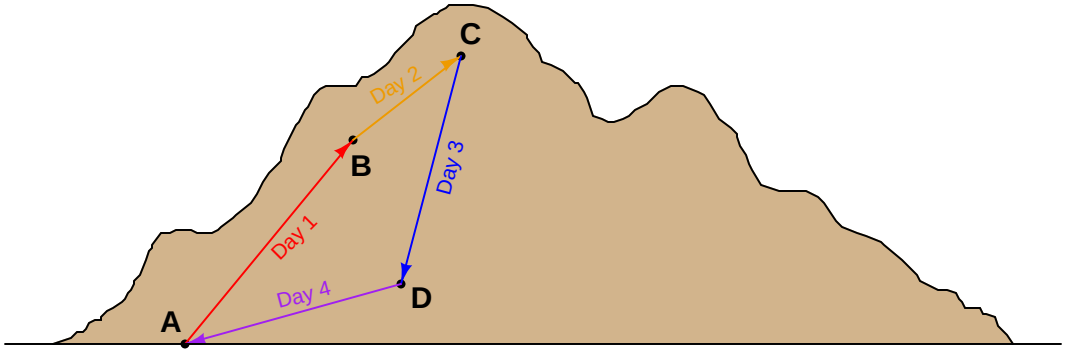
For both series and parallel circuits, total power dissipated by all load devices is equal to the total power delivered by all source devices. The configuration of a circuit is irrelevant to the balance between power supplied and power lost, because this balance is an expression of the Law of Energy Conservation.

4.7 Kirchhoff's Laws

Two extremely important principles in electric circuits were codified by Gustav Robert Kirchhoff in the year 1847, known as *Kirchhoff's Laws*. His two laws refer to voltages and currents in electric circuits, respectively.

Kirchhoff's Voltage Law states that the algebraic sum of all voltages in a closed loop is equal to zero. Another way to state this law is to say that for every rise in potential there must be an equal fall, if we begin at any point in a circuit and travel in a loop back to that same starting point.

An analogy for visualizing Kirchhoff's Voltage Law is hiking up a mountain. Suppose we start at the base of a mountain and hike to an altitude of 5000 feet to set up camp for an overnight stay. Then, the next day we set off from camp and hike farther up another 3500 feet. Deciding we've climbed high enough for two days, we set up camp again and stay the night. The next day we hike down 6200 feet to a third location and camp once gain. On the fourth day we hike back to our original starting point at the base of the mountain. We can summarize our hiking adventure as a series of rises and falls like this:

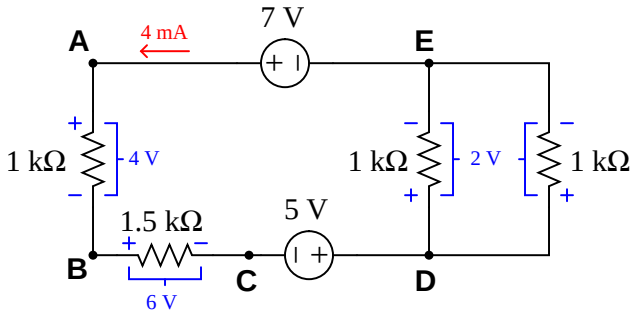


Day	Path	Altitude gain/loss
Day 1	A to B	+5000 feet
Day 2	B to C	+3500 feet
Day 3	C to D	−6200 feet
Day 4	D to A	−2300 feet
(Total)	ABCDA	0 feet

Of course, no one would tell their friends they spent *four days hiking a total altitude of 0 feet*, so people generally speak in terms of the *highest* point reached: in this case 8500 feet. However, if we track each day's gain or loss in algebraic terms (maintaining the mathematical sign, either positive or negative), we see that the end sum is zero (and indeed *must always be zero*) if we finish at our starting point.

If we view this scenario from the perspective of potential energy as we lift a constant mass from point to point, we would conclude that we were doing work on that mass (i.e. investing energy in it by lifting it higher) on days 1 and 2, but letting the mass do work on us (i.e. releasing energy by lowering it) on days 3 and 4. After the four-day hike, the net potential energy imparted to the mass is zero, because it ends up at the exact same altitude it started at.

Let's apply this principle to a real circuit, where total current and all voltage drops have already been calculated for us:



If we trace a path ABCDEA, we see that the algebraic voltage sum in this loop is zero:

Path	Voltage gain/loss
A to B	− 4 volts
B to C	− 6 volts
C to D	+ 5 volts
D to E	− 2 volts
E to A	+ 7 volts
ABCDEA	0 volts

We can even trace a path that does not follow the circuit conductors or include all components, such as EDCBE, and we will see that the algebraic sum of all voltages is still zero:

Path	Voltage gain/loss
E to D	+ 2 volts
D to C	− 5 volts
C to B	+ 6 volts
B to E	− 3 volts
EDCBE	0 volts

Kirchhoff's Voltage Law is often a difficult subject for students, precisely because voltage itself is a difficult concept to grasp. Remember that there is no such thing as voltage at a single point; rather, voltage exists only as a *differential* quantity. To intelligently speak of voltage, we must refer to either a *loss* or *gain* of potential between **two points**.

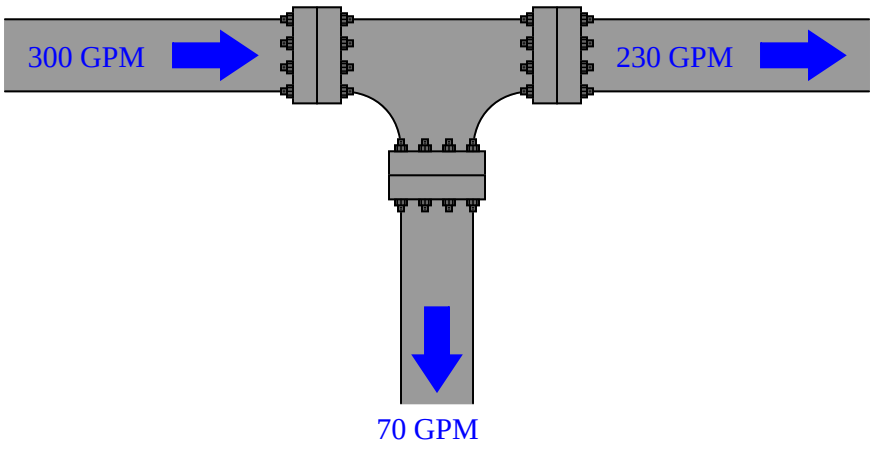
Our analogy of altitude on a mountain is particularly apt. We cannot intelligently speak of some point on the mountain as having a specific altitude unless we assume a point of reference to measure from. If we say the mountain summit is 9200 feet high, we usually mean 9200 feet *higher than sea level*, with the level of the sea being our common reference point. However, our hiking adventure where we climbed 8500 feet in two days did not imply that we climbed to an absolute altitude of 8500 feet above sea level. Since I never specified the sea-level altitude at the base of the mountain,

it is impossible to calculate our absolute altitude at the end of day 2. All you can tell from the data given is that we climbed 8500 feet *above* the mountain base, wherever that happens to be with reference to sea level.

So it is with electrical voltage as well: most circuits have a point labeled as *ground* where all other voltages are referenced. In DC-powered circuits, this ground point is often the negative pole of the DC power source[12]. Voltage is fundamentally a quantity relative between two points: a measure of how much potential has *increased* or *decreased* moving from one point to another.

Kirchhoff's Current Law is a much easier concept to grasp. This law states that the algebraic sum of all currents at a junction point (called a *node*) is equal to zero. Another way to state this law is to say that for every electron entering a node, one must exit somewhere.

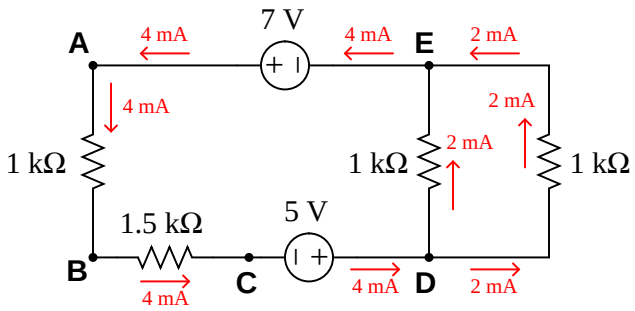
An analogy for visualizing Kirchhoff's Current Law is water flowing into and out of a "tee" fitting:



So long as there are no leaks in this piping system, every drop of water entering the tee must be balanced by a drop exiting the tee. For there to be a continuous mis-match between flow rates would imply a violation of the Law of Mass Conservation.

[12] But not always! There do exist positive-ground systems, particularly in telephone circuits and in some early automobile electrical systems.

Let's apply this principle to a real circuit, where all currents have been calculated for us:



At nodes where just two wires connect (such as points A, B, and C), the amount of current going in to the node exactly equals the amount of current going out (4 mA, in each case). At nodes where three wires join (such as points D and E), we see one large current and two smaller currents (one 4 mA current versus two 2 mA currents), with the directions such that the sum of the two smaller currents form the larger current.

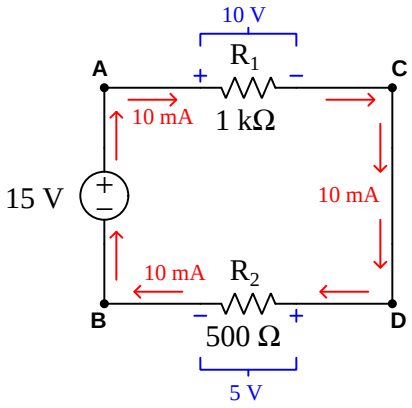
Just as the balance of water flow rates into and out of a piping "tee" is a consequence of the Law of Mass Conservation, the balance of electric currents flowing into and out of a circuit junction is a consequence of the Law of *Charge* Conservation, another fundamental conservation law in physics.

4.8 Circuit fault analysis

Perhaps the most valuable skill an instrument technician can possess is the ability to efficiently diagnose malfunctioning systems: determining in as short a time as possible the cause of a system's malfunction. Since most instrumentation and control systems are based on electricity, a solid understanding of electrical faults is the foundation of this skill set. In this section we will explore the two basic types of electrical faults (*opens* and *shorts*) and analyze their respective effects in DC circuits.

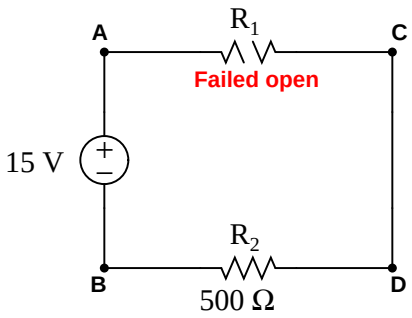
An *open* is an electrical fault whereby the pathway for electrical current is broken, thus preventing the passage of current. A *short* is an electrical fault whereby two points in a circuit that are supposed to be separated are joined together by a conductive pathway. It should be noted that this definition for the word "short" both is technical and specific. This is important to understand, as many people tend to use the word "short" to refer to any electrical problem in general. In technical parlance, a "short" fault is the exact opposite of an "open" fault, and should never be confused one for the other.

Let us examine the effects of both faults in a simple two-resistor DC circuit. We will begin with an analysis of the circuit in a healthy condition, showing all values of voltage and current:



Being a series circuit, the current is the same through all components while the two loads' voltage drops add to equal the source voltage.

Now suppose resistor R_1 fails open. We know that a continuous electric current is able to exist only where there is a continuous path for that current. If a component fails open, the continuity of the circuit will be broken, and current must halt. Since this is a series circuit, if current halts in one location it must likewise halt in all locations:

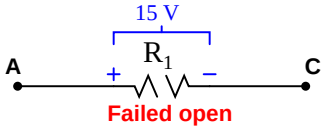


We would still expect to measure 15 volts between test points A and B because it is the nature of a voltage source to maintain a constant voltage across its terminals (i.e. to maintain the same amount of energy difference per unit charge from one side to the other). We may apply Ohm's Law to calculate the voltage across the healthy resistor R_2: since $V = IR$ and $I = 0$, we may safely conclude that $V = 0$ for resistor R_2.

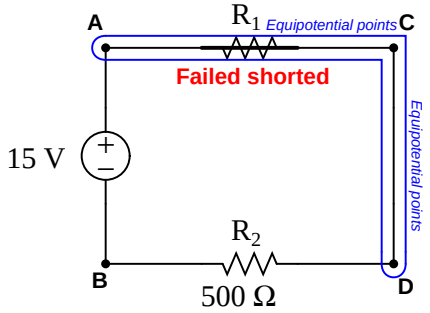
Unfortunately, we cannot apply Ohm's Law to the failed resistor R_1 since the resistance of any failed-open component is infinite. Infinitudes do not lend themselves to arithmetic calculations (how much is zero times infinity?), and so we must find some other way to determine the voltage dropped across the open R_1. Here we find that Kirchhoff's Voltage Law applies very nicely: if we know that the algebraic sum of all voltages in a loop must equal zero, and we know all but one of the voltages in the loop BACDB, we may calculate the last voltage by simple addition and subtraction:

Path	Voltage gain/loss	Reason
B to A	+ 15 volts	Source voltage value
A to C	**−15 volts**	Necessary to satisfy KVL
C to D	0 volts	These points are equipotential
D to B	0 volts	Ohm's Law: $V = 0$ amps $\times$ 500 ohms
BACDB	0 volts	KVL sum = 0

Clearly, the voltage between points A and C (across the failed-open R_1) must be −15 volts in order to satisfy Kirchhoff's Voltage Law. The negative sign of V_{AC} tells us that point C must be at a lesser potential than point A (i.e. A is positive and C is negative).



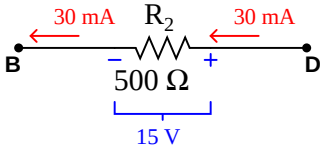
Let us suppose now resistor R_1 fails shorted. This means its two terminals are now directly connected to each other rather than being separated by 1000 ohms of electrical resistance. In other words, the two terminals of resistor R_1 have been made *equipotential* by the shorted fault. By extension, this forces points A and C to be equipotential as well. Since points C and D are already equipotential to each other by virtue of the wire connecting them, points A through D must now be equipotential.



If points A through D are all equipotential, it means the right-hand terminal of resistor R_2 is directly connected to the positive terminal of the 15 volt source. The left-hand terminal of R_2 is already equipotential with the negative terminal of the source. This means the shorted fault at R_1 has placed the full source potential across resistor R_2. An analysis using Kirchhoff's Law confirms this:

Path	Voltage gain/loss	Reason
B to A	+ 15 volts	Source voltage value
A to C	0 volts	Shorting points makes them equipotential
C to D	0 volts	These points are equipotential
D to B	**−15 volts**	Necessary to satisfy KVL
BACDB	0 volts	KVL sum = 0

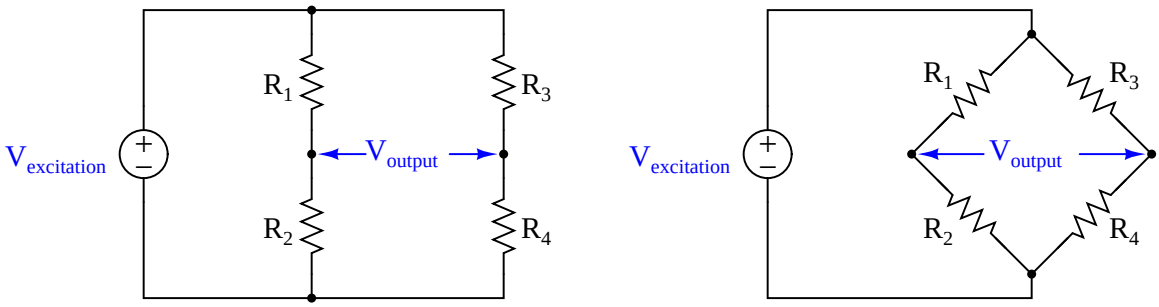
The negative sign of V_{DB} tells us that point B must be at a lesser potential than point D (i.e. D is positive and B is negative). Calculating the amount of current in this circuit is now a simple matter of applying Ohm's Law to resistor R_2. Given a 15 volt potential across it and 500 ohms of resistance, the current is equal to 30 milliamps:



All DC circuit fault analysis reduces to these simple principles: *open* faults directly affect current by interrupting the continuity of the circuit, while *shorted* faults directly affect voltage by making points equipotential to each other that were not equipotential before. The rest is merely applying Kirchhoff's and Ohm's Laws to determine consequences of the fault throughout the circuit.

4.9 Bridge circuits

A *bridge* circuit is basically a pair of voltage dividers where the circuit output is taken as the difference in potential between the two dividers. Bridge circuits may be drawn in schematic form in an H-shape or in a diamond shape, although the diamond configuration is more common:

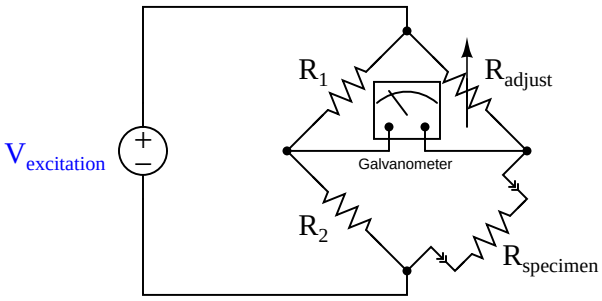


The voltage source powering the bridge circuit is called the *excitation* source. This source may be DC or AC depending on the application of the bridge circuit. The components comprising the bridge need not be resistors, either: capacitors, inductors, lengths of wire, sensing elements, and other component forms are possible, depending on the application.

Two major applications exist for bridge circuits, which will be explained in the following subsections.

4.9.1 Component measurement

Bridge circuits may be used to test components. In this capacity, one of the "arms" of the bridge circuit is comprised of the component under test, while at least one of the other "arms" is made adjustable. The common *Wheatstone bridge* circuit for resistance measurement is shown here:

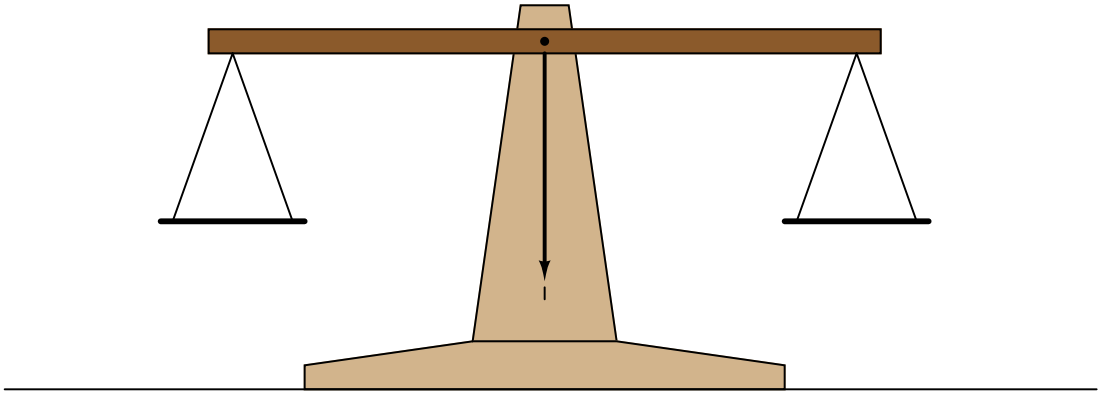


Fixed resistors R_1 and R_2 are of precisely known value and high precision. Variable resistor R_{adjust} has a labeled knob allowing for a person to adjust and read its value to a high degree of precision. When the ratio of the variable resistance to the specimen resistance equals the ratio of the two fixed resistors, the sensitive galvanometer will register exactly zero volts regardless of the excitation source's value. This is called a *balanced* condition for the bridge circuit:

$$\frac{R_1}{R_2} = \frac{R_{adjust}}{R_{specimen}}$$

When the two resistance ratios are equal, the voltage drops across the respective resistances will also be equal. Kirchhoff's Voltage Law declares that the voltage differential between two equal and opposite voltage drops must be zero, accounting for the meter's indication of balance.

It would not be inappropriate to relate this to the operation of a laboratory balance-beam scale, comparing a specimen of unknown mass against a set of known masses. In either case, the instrument is merely comparing an unknown quantity against an (adjustable) known quantity, indicating a condition of equality between the two:

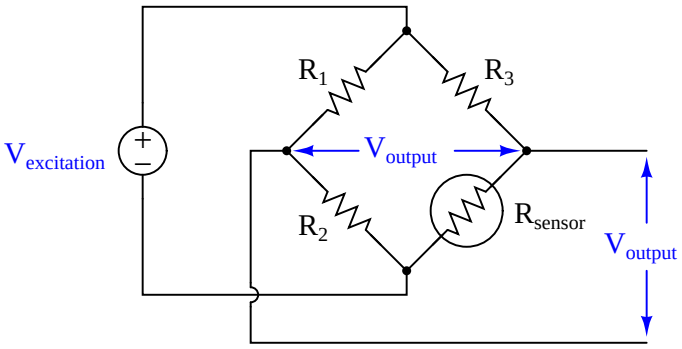


Many legacy instruments were designed around the concept of a *self-balancing* bridge circuit, where an electric servo motor drove a potentiometer to achieve a balanced condition against the voltage produced by some process sensor. Analog electronic paper chart recorders often used this principle. Almost all pneumatic process instruments use this principle to translate the force of a sensing element into a variable air pressure.

Modern bridge circuits are mostly used in laboratories for extremely precise component measurements. Very rarely will you encounter a Wheatstone bridge circuit used in the process industries.

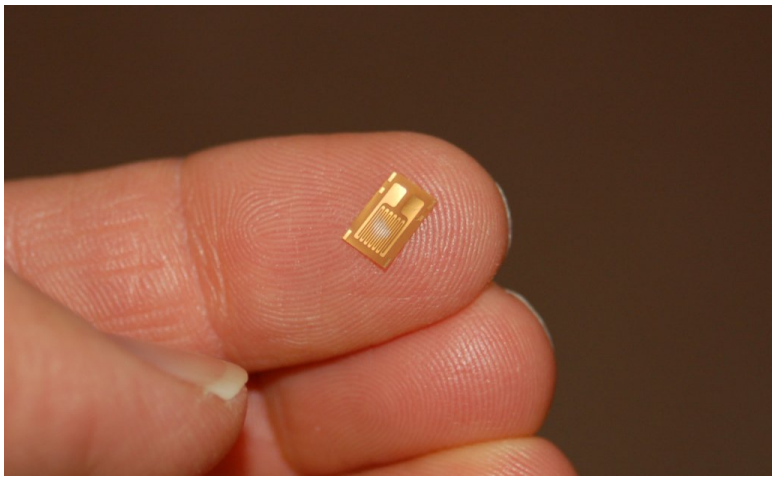
4.9.2 Sensor signal conditioning

A different application for bridge circuits is to convert the output of an electrical sensor into a voltage signal representing some physical measurement. This is by far the most popular use of bridge measurement circuits in industry, and here we see the same circuit used in an entirely different manner from that of the balanced Wheatstone bridge circuit.

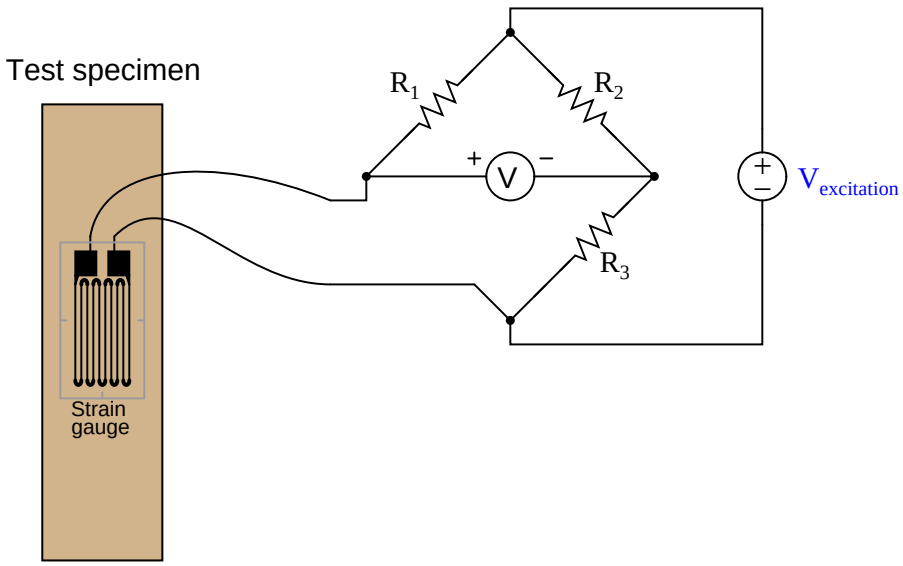


Here, the bridge will be balanced only when R_{sensor} is at one particular resistance value. Unlike the Wheatstone bridge, which serves to measure a component's value when the circuit is balanced, this bridge circuit will probably spend most of its life in an unbalanced condition. The output voltage changes as a function of sensor resistance, which makes that voltage a reflection of the sensor's physical condition. In the above circuit, we see that the output voltage increases (positive on the top wire, negative on the bottom wire) as the resistance of R_{sensor} increases.

One of the most common applications for this kind of bridge circuit is in strain measurement, where the mechanical strain of an object is converted into an electrical signal. The sensor used here is a device known as a *strain gauge*: a folded wire designed to stretch and compress with the object under test, altering its electrical resistance accordingly. Strain gauges are typically quite small, as shown by this photograph:



Strain gauges are useful when bonded to metal specimens, providing a means of electrically sensing the strain ("stretching" or "compressing" of that specimen). The following bridge circuit is a typical application for a strain gauge:



When the specimen is stretched along its long axis, the metal wires in the strain gauge stretch with it, increasing their length and decreasing their cross-sectional area, both of which work to

increase the wire's electrical resistance. This stretching is microscopic in scale, but the resistance change is measurable and repeatable within the specimen's elastic limit. In the above circuit example, stretching the specimen will cause the voltmeter to read upscale (as defined by the polarity marks). Compressing the specimen along its long axis has the opposite effect, decreasing the strain gauge resistance and driving the meter downscale.

Strain gauges are used to precisely measure the strain (stretching or compressing motion) of mechanical elements. One application for strain gauges is the measurement of strain on machinery components, such as the frame components of an automobile or airplane undergoing design development testing. Another application is in the measurement of force in a device called a *load cell*. A "load cell" is comprised of one or more strain gauges bonded to the surface of a metal structure having precisely known elastic properties. This metal structure will stretch and compress very precisely with applied force, as though it were an extremely stiff spring. The strain gauges bonded to this structure measure the strain, translating applied force into electrical resistance changes.

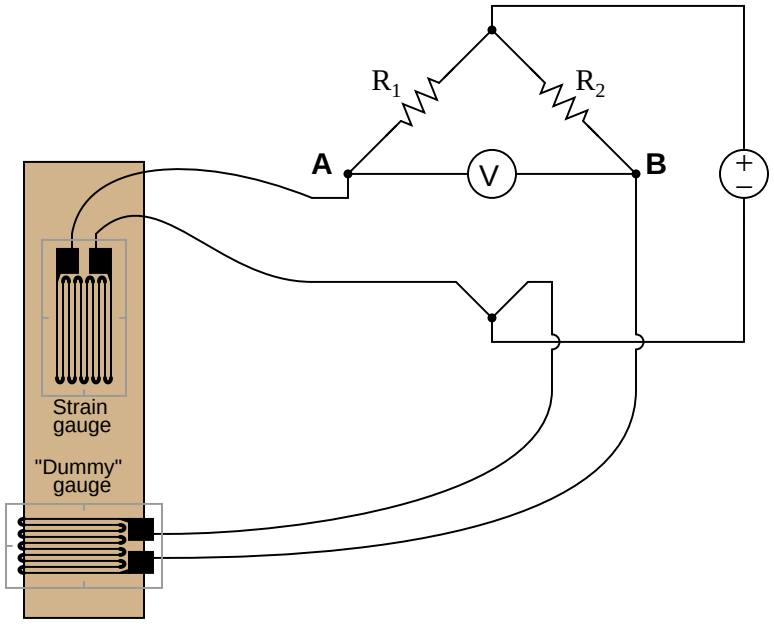
You can see what a load cell looks like in the following photograph:



Strain gauges are not the only dynamic element applicable to bridge circuits. In fact, any resistance-based sensor may be used in a bridge circuit to translate a physical measurement into an electrical (voltage) signal. Thermistors (changing resistance with temperature) and photocells (changing resistance with light exposure) are just two alternatives to strain gauges.

It should be noted that the amount of voltage output by this bridge circuit depends both on the amount of resistance change of the sensor *and* the value of the excitation source. This dependency on source voltage value is a major difference between a sensing bridge circuit and a Wheatstone (balanced) bridge circuit. In a perfectly balanced bridge, the excitation voltage is irrelevant: the output voltage is zero no matter what source voltage value you use. In an unbalanced bridge circuit, however, source voltage value matters! For this reason, these bridge circuits are often rated in terms of how many millivolts of output they produce *per volt of excitation* per unit of physical measurement (microns of strain, newtons of stress, etc.).

An interesting feature of a sensing bridge circuit is its ability to cancel out unwanted variables. In the case of a strain gauge, for example, mechanical strain is not the only variable affecting gauge resistance. Temperature also affects gauge resistance. Since we do not wish our strain gauge to also act as a thermometer (which would make measurements very uncertain – how would we differentiate the effects of changing temperature from the effects of changing strain?), we must find some way to nullify resistance changes due solely to temperature, such that our bridge circuit will respond only to changes in strain. The solution is to creatively use a "dummy" strain gauge as another arm of the bridge:

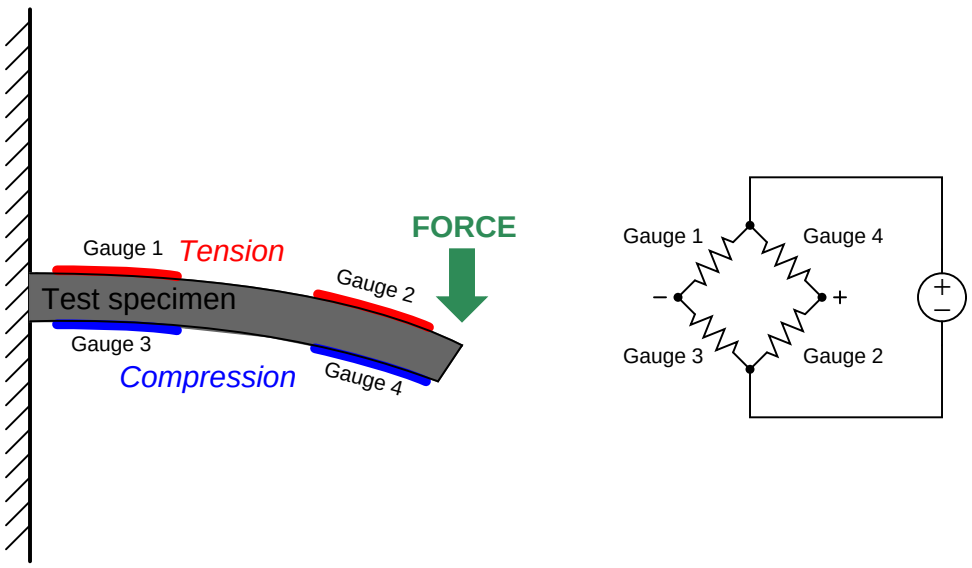


The "dummy" gauge is attached to the specimen in such a way that it maintains the same temperature as the active strain gauge, yet experiences no strain. Thus, any *difference* in gauge resistances must be due solely to specimen strain. The differential nature of the bridge circuit naturally translates the differential resistance of the two gauges into one voltage signal representing strain.

If thermistors are used instead of strain gauges, this circuit becomes a differential temperature sensor. Differential temperature sensing circuits are used in solar heating control systems, to detect when the solar collector is hotter than the room or heat storage mass being heated.

Sensing bridge circuits may have more than one active "arm" as well. The examples you have seen so far in this section have all been *quarter-active* bridge circuits. It is possible, however, to incorporate more than one sensor into the same bridge circuit. So long as the sensors' resistance changes are coordinated, their combined effect will be to increase the sensitivity (and often the linearity as well) of the measurement.

For example, *full-active bridge* circuits are sometimes built out of four strain gauges, where each strain gauge comprises one arm of the bridge. Two of the strain gauges must compress and the other two must stretch under the application of the same mechanical force, in order that the bridge will become unbalanced with strain:

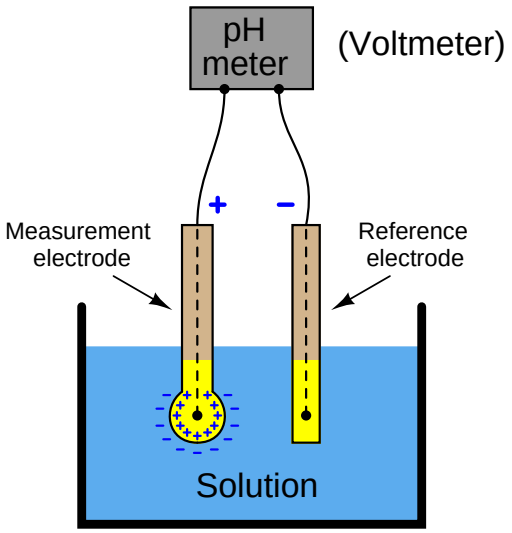


Not only does a full-active bridge circuit provide greater sensitivity and linearity than a quarter-active bridge, but it also *naturally* provides temperature compensation without the need for "dummy" strain gauges, since the resistances of all four strain gauges will change by the same proportion if the specimen temperature changes.

4.10 Null-balance voltage measurement

A number of different sensors used in instrumentation generate DC voltage signals proportional to the process variable of interest. We call such sensors *potentiometric*, which literally means "voltage-measuring". Thermocouples are one type of potentiometric sensor, used to measure temperature. Photodiodes are another, used to measure light intensity. Glass pH electrodes are yet another, used to measure the hydrogen ion activity in a liquid solution. It should be obvious that accurate voltage measurement is critical for any instrument based on a potentiometric sensor, for if our measurement of that sensor's output voltage is not accurate, we will surely suffer inaccurate measurement of any process variable proportional to that voltage (e.g. temperature, light, pH).

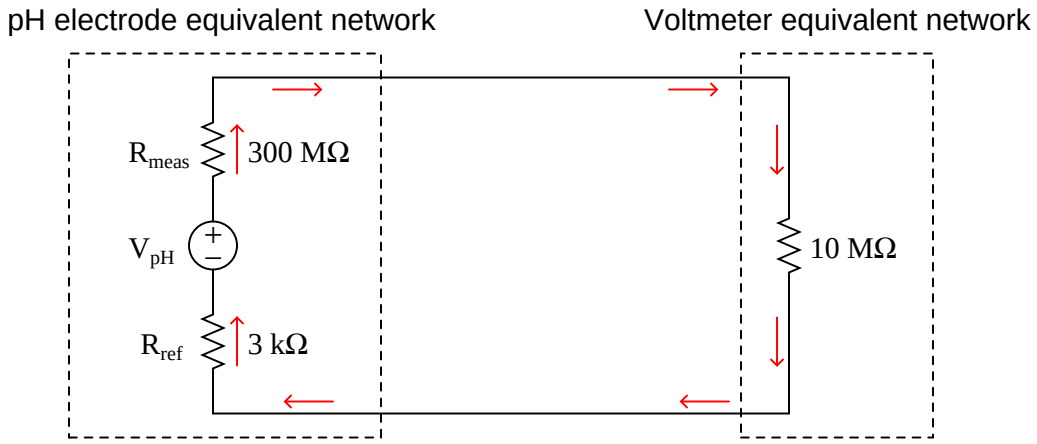
One common obstacle to accurate sensor voltage measurement is the internal resistance of the sensor itself. We will explore this concept by way of a practical example: trying to measure the voltage output by a pH electrode pair using a standard digital voltmeter. A pictorial diagram shows the basic concept, where a voltmeter is connected to a pH electrode pair immersed in a liquid solution:



Hydrogen ions within the liquid solution penetrate the round glass bulb of the measurement electrode, generating a potential difference approximately equal to 59 millivolts per pH unit of deviation from 7 pH (neutral). The reference electrode serves the simple purpose of completing the electrical circuit from the voltmeter's terminals to both sides of the glass bulb (inside and outside).

What should be an elementary task is complicated by the fact that the glass bulb of the measurement electrode has an incredibly high electrical resistance, typically on the order of *hundreds* of mega-ohms. When connected to a common digital multimeter having an input resistance in the order of *tens* of mega-ohms, the voltmeter acts as a rather "heavy" electrical load which causes the measured voltage to be far less than what the glass electrode is actually producing.

If we sketch an equivalent electrical schematic of these components, the problem becomes more evident. Red arrows in this schematic depict the flow of electrical current (conventional notation):



Only a small fraction of the glass electrode's voltage (V_{pH}) will actually be seen at the voltmeter's terminals due to this loading effect. We may treat the voltmeter's internal resistance of 10 MΩ as one resistance in a voltage divider network, the two probe resistances being the other two divider resistances:

$$V_{meter} = V_{pH} \left(\frac{10 \text{ M}\Omega}{300 \text{ M}\Omega + 10 \text{ M}\Omega + 3 \text{ k}\Omega} \right)$$

Supposing the pH-sensing glass bulb outputs 100 millivolts, the voltmeter in this circuit would only register a reading of 3.226 millivolts: just a *few percent* of the actual sensor's potentiometric output. While this is a rather extreme example, it should be clear to see that any potentiometric circuit of the same form will suffer some degree of measurement inaccuracy due to this effect – the only question being how much error.

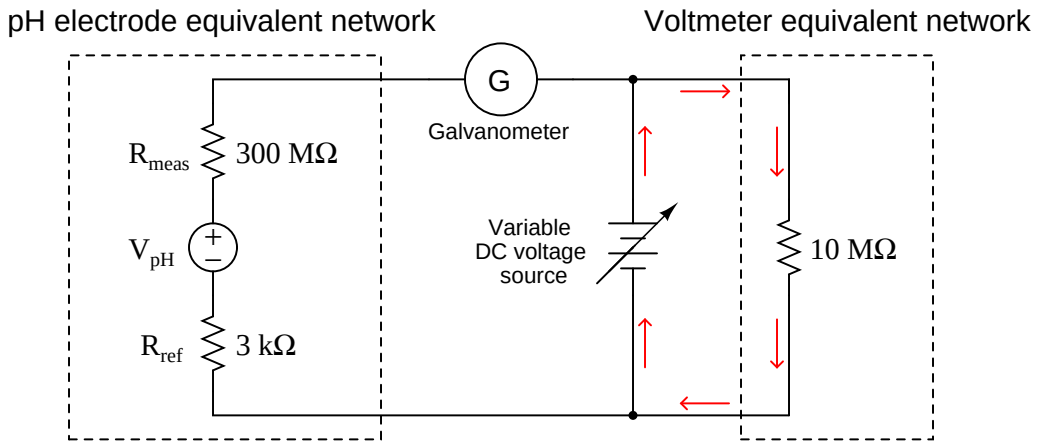
Lying at the heart of this problem is the fact that voltmeters necessarily draw some electric current in the act of measuring a voltage. It is this current draw, no matter how slight, that causes a voltmeter to register something other than a perfect facsimile of the sensor's voltage signal. The solution to this problem, then, is to minimize or eliminate this current draw. In other words, we need our voltmeter to have as much internal resistance as possible (ideally, an *infinite* amount of internal resistance).

Modern field-effect transistor amplifier circuits go a long way toward addressing this problem by allowing us to manufacture voltmeters having internal resistances in the *trillions* of ohms. So long as the voltmeter's internal resistance far overshadows (i.e. "swamps") the signal source's resistance, loading errors will be held to a minimum.

In the days before high-resistance semiconductor amplifier circuits, special voltmeters called *vacuum-tube voltmeters* (VTVMs) were used whenever voltages needed to be measured from high-resistance potentiometric sensors.

Prior to the advent of electronic vacuum tubes, though, it was impossible to mitigate the problem of voltmeter loading simply by selecting a better-quality voltmeter. Un-amplified voltmeters rely on the passage of a small current from the source under test to drive their indicating mechanisms. Without this small current drawn from the circuit, the voltmeter simply would not function at all. How then did early experimenters and electrical metrologists overcome this problem of loading?

An ingenious solution to the problem of voltmeter loading is the so-called *null-balance* method of voltage measurement. This technique added two components to the measurement circuit: a highly sensitive ammeter called a *galvanometer* and a variable DC voltage source. Using our pH measurement circuit as an example, a null-balance arrangement would look something like this:



Operation of this circuit follows these two steps:

- Adjust the variable DC voltage source until the galvanometer registers exactly zero (i.e. no current)
- Read the voltmeter to see what the pH sensor's voltage is

So long as the galvanometer registers zero (a "null" condition), there will be no electric current passing through the large resistances of the pH sensor's electrodes because the pH sensor's voltage is perfectly balanced against the variable supply's voltage. With no current passing through those high resistances, they will drop no voltage whatsoever. Thus, V_{pH} must be equal to the voltage of the variable DC source, which the voltmeter registers accurately because its current requirements are met by the variable source and not the pH sensor. The only way this measurement technique can fail in its objective is if the galvanometer is not able to precisely detect a condition of zero current. So long as the galvanometer faithfully tells us when we have reached a condition of zero current, we may measure the voltage of the pH sensor using *any* DC voltmeter regardless of its internal resistance.

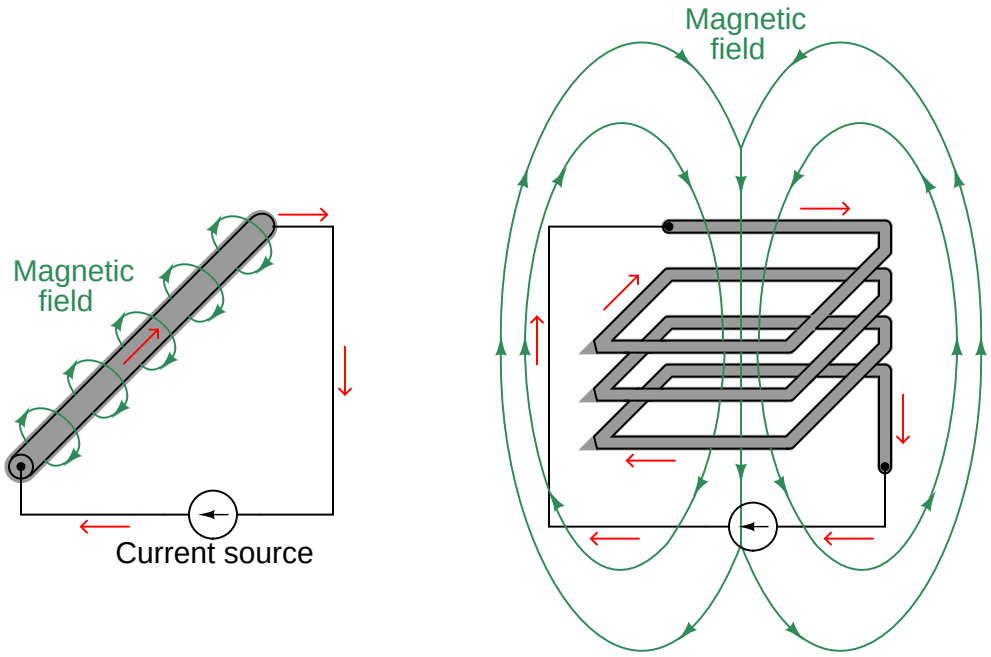
Special null-balance voltmeter instruments were manufactured with precision variable voltage sources built into them, called *differential voltmeters*. One such instrument was the Fluke model 801, a gold-plated version of which is shown here in Fluke's museum of measurement:



Note the center-zero analog meter on the face of this instrument, performing the function of the sensitive galvanometer in our schematic diagram. A set of five knobs oriented vertically on the face of this instrument, each one showing one digit of a 5-digit number, adjusted the DC voltage output by the internal voltage source. When the "null" meter registered zero, it meant the voltage of the source or circuit under test was precisely equal to the voltage dialed up by these five knobs. Differential voltmeters such as the Fluke 801 used amplifier circuits to make this "null" detector ultra-sensitive, in order to achieve the most accurate condition of balance between the variable DC voltage source and the source under test possible.

4.11 Electromagnetism

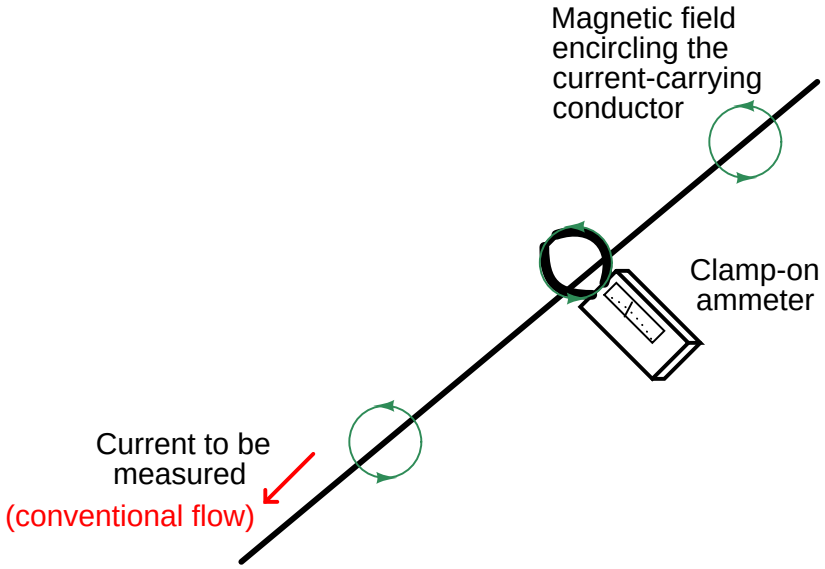
The fundamental principle of *electromagnetism* is that an electric current will create a magnetic field at right angles to the direction of the current. If the electric current travels in a straight path, the lines of magnetic flux will form concentric circles around that path. If the electric current travels in a circular path (i.e. through a loop or coil of wire), the magnetic lines of flux will form straight lines down the center of the coil, wrapping around at the ends to form a complete loop of its own:



Magnetic field strength is directly proportional to the amount of current in the conductor (and also directly proportional to the number of "turns" in a coiled wire), such that the unit of measurement[13] for magnetic field strength is the *amp-turn*.

[13] Both in the British system of measurement *and* the SI metric system of measurement! The older metric system (called "CGS" for Centimeter-Gram-Second) had a special unit of measurement called the *Gilbert* for expressing magnetic field strength, with 1 Gilbert (Gb) equal to 0.7958 Amp-turns (At).

The directly proportional relationship between current intensity and magnetic field strength is exploited by *clamp-on ammeters*, which are able to measure electric current without the need for contact with the conductor:



The following photograph shows how particles of steel align with the circular magnetic field surrounding one of the electric cables of an electric arc welder on a job site:

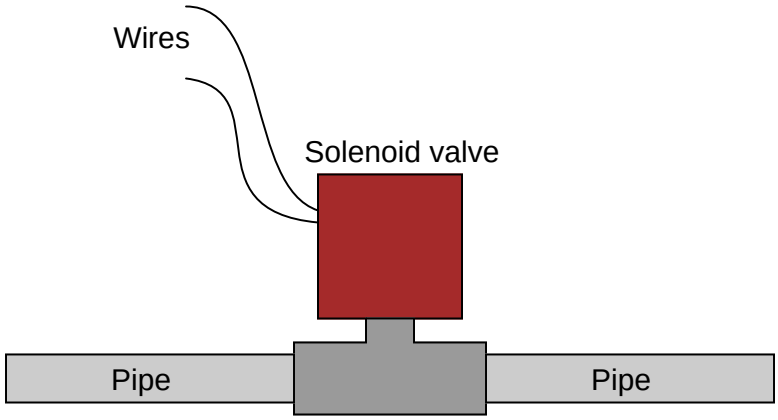


Very strong magnetic fields may be generated with wire *coils*, since the magnetic fields surrounding each "turn" of wire in a coil tend to overlap constructively, supporting one another to form a stronger total field. The magnetic field from a wire coil may be so strong, in fact, that it is useful for creating an attractive force upon a ferrous[14] object (called an *armature*) strong enough to move mechanisms. This arrangement of a wire coil and an iron armature is called a *solenoid*[15].

[14]The term "ferrous" simply refers to any substance containing the element *iron*. This includes steel, which is a combination of iron and carbon.

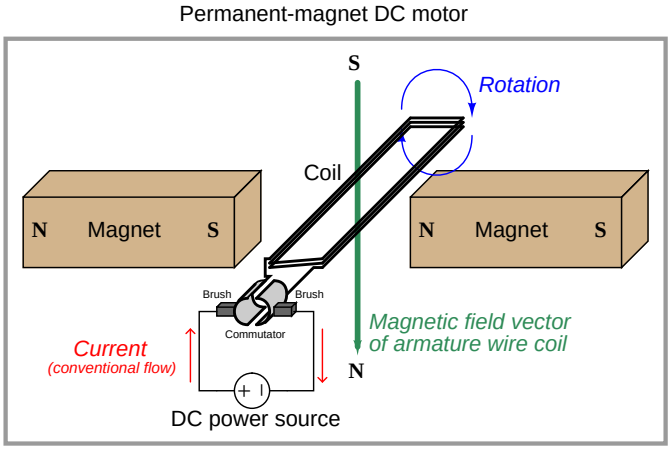
[15]The word "solenoid" may also be used to describe a wire coil with no armature, but the more common industrial use of the word refers to the complete arrangement of coil and movable armature.

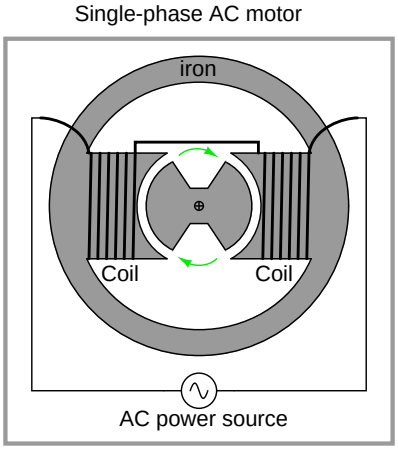
A practical example of a solenoid is a *solenoid valve*: a mechanical valve opened and/or closed by the application of electric current through the coil:



When the coil is energized by an external source of electric current, the magnetic field attracts the movable armature, thereby actuating the valve. A spring typically returns the valve mechanism back to its original position upon de-energization of the coil.

Another very practical application of electromagnetism are *electric motors*, which use magnetic fields generated by wire coils to twist a mechanical shaft and perform useful work.





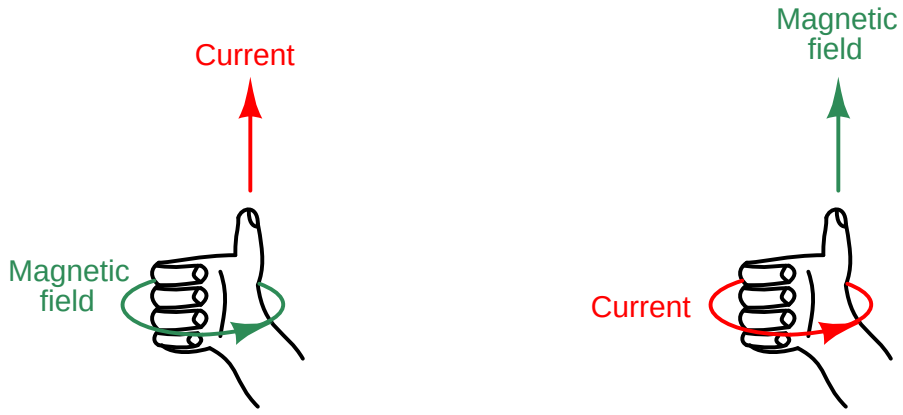
In the DC motor design, electric current creates a magnetic field in the armature coil which reacts against the magnetic fields of the permanent magnets to twist the armature about its axis. Every half-turn, the brushes break and re-make contact with the commutator bars, reversing the direction of current through the armature coil to keep it spinning in the same direction.

In the AC motor design, alternating current (continually reversing direction many times per second) creates an alternating magnetic field in a set of stationary wire coils called *stator* windings.

This alternating magnetic field causes a rotor to spin at speed proportional to the frequency of the magnetic field's reversals.

Both electric current and magnetic field lines are *vectors*, having both magnitude and direction. As we have seen already, there is a perpendicular relationship between these two vectors. This relationship may be visualized by a simple rule called the *right-hand rule*, whereby the fingers and thumb of a human right hand represent the vector orientations of current and magnetism (or vice-versa). Using the right-hand rule, digits representing current direction assume the use of *conventional flow* rather than electron flow[16], while digits representing magnetism point in the direction of "North."

To use this rule, curl the four fingers of your right hand such that they point toward the palm of that hand, and extend your right thumb so that it points perpendicularly to your curled fingers. Your right hand should look like this:

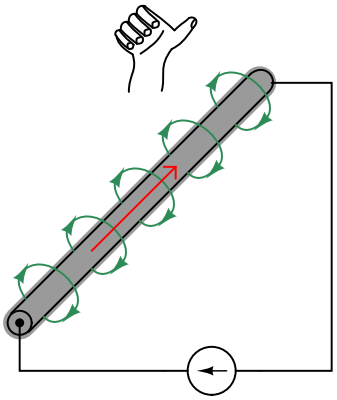


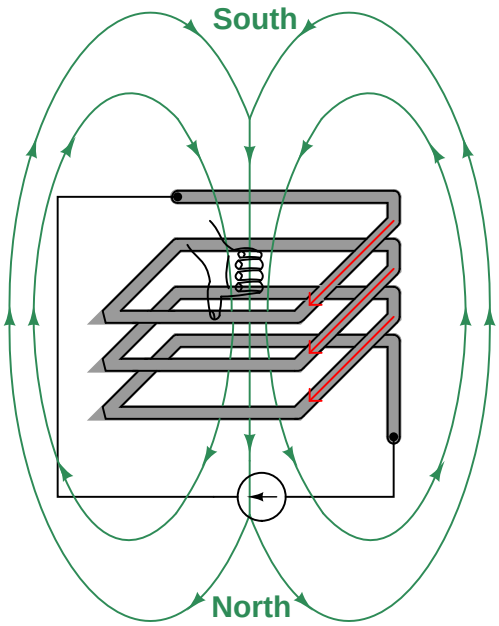
Your curled fingers will represent one set of vectors, while your thumb points in the direction of the other. Whether the fingers represent current and the thumb magnetism, or whether the fingers represent magnetism and the thumb current, is irrelevant: the rule works both ways.

[16] There is also a left-hand rule for fans of electron flow, but in this book I will default to conventional flow. For a more complete discussion on this matter, see section 4.2.1 beginning on page 301.

This flexibility makes the right-hand rule easy to apply to different situations such as these:

Thumb represents current vector
Fingers represent magnetic field vectors

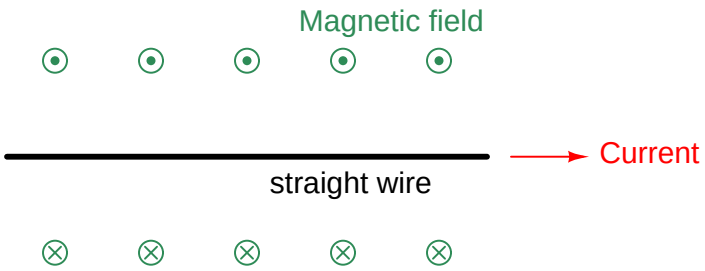




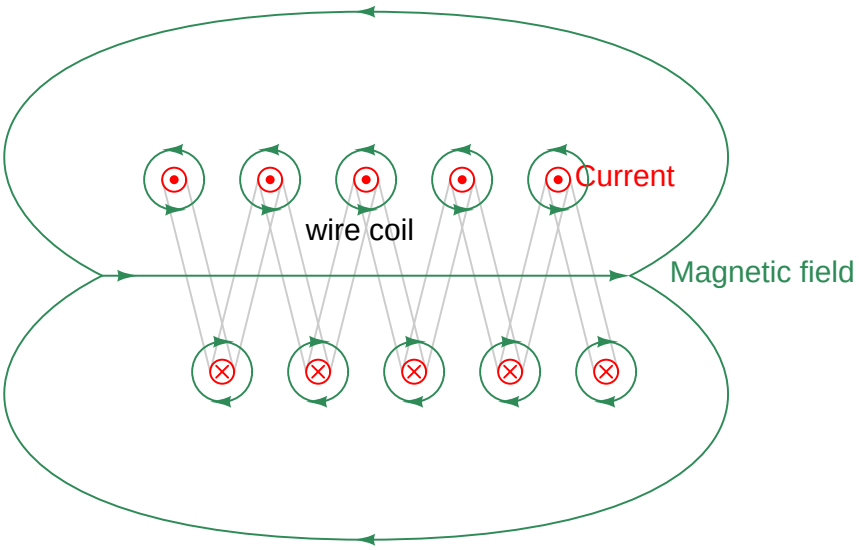
Fingers represent current vectors
Thumb represents magnetic field vector

Note: red arrows point in the direction of conventional current flow

Physicists have devised a convention for clearly illustrating the directions of perpendicular vectors (arrows) without resorting to illustrations drawn in 3-dimensional perspective. The following two-dimensional illustration shows the magnetic field surrounding a straight, current-carrying wire. The magnetic field, of course, encircles the wire. This is shown by the alternating "dot" and "cross" symbols above and below the wire, representing arrow heads (circles with dots) coming "out" of the page directly at the reader, and representing arrow tails (circles with crosses) headed straight into the page away from the reader:



The same notation may be used to show the perpendicular relationship between current and magnetic flux lines for a coiled conductor. Here, the arrow "tips" and "tails" represent current (conventional flow) entering and exiting the page, while the horizontal arrow represents magnetic field direction:



Note how the individual magnetic fields surrounding each wire in the coil all have their arrows pointing to the right in the coil's interior, and to the left at the coil's exterior. This shows how the individual magnetic loops constructively add to generate a large magnetic field through the center of the coil, looping around back to the other end of the coil.

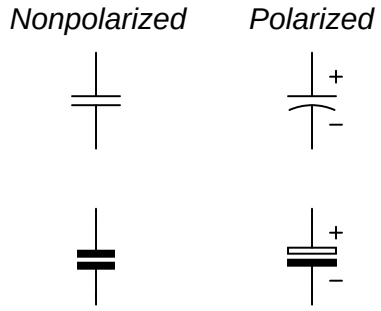
4.12 Capacitors

Any two electrical conductors separated by an insulating medium possess the characteristic called *capacitance*: the ability to store energy in the form of an electric field created by a voltage between those two conductors. Capacitance is symbolized by the capital letter C and is measured in the unit of the *Farad* (F). The relationship between capacitance, stored electric charge (Q), and voltage (V) is as follows:

$$Q = CV$$

For example, a capacitance having a value of 33 microfarads charged to a voltage of 5 volts would store an electric charge of 165 microcoulombs.

Capacitors are devices expressly designed and manufactured to possess capacitance. They are constructed of a "sandwich" of conductive plates separated by an insulating *dielectric*. Capacitors have maximum voltage ratings as well as capacitance ratings. Here are some schematic symbols for capacitors:



Although any two conductors separated from each other by an insulating medium possess capacitance, we may optimize the amount of capacitance by situating the conductors in a manner yielding more electric flux for any given amount of applied voltage. Placing the two conductors closer together (thus creating a more intense electric field between them, measured in *volts per inch*) is one way to increase capacitance. Increasing the overlapping area of the conductors (thus providing more area for parallel electric flux lines to form) is another way. Inserting a strongly dielectric (i.e. an exceptionally good insulating) material between the conductors is a third way to increase capacitance.

These three factors may be expressed in a single mathematical formula for capacitance:

$$C = \frac{\epsilon A}{d}$$

Where,

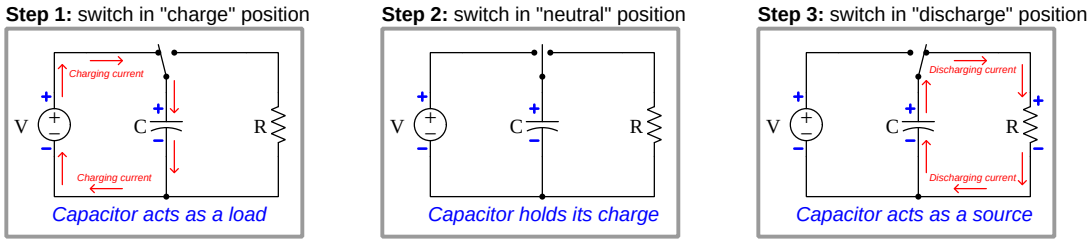
C = Capacitance in Farads

A = Overlapping area of conductors in square meters

ϵ = Permittivity of the dielectric substance

d = Distance separating the conductors in meters

Capacitance is a non-dissipative quantity. Unlike resistance, a pure capacitance does not dissipate energy in the form of heat; rather, it stores and releases energy from and to the rest of the circuit. We may illustrate the energy-storing behavior of a capacitor by this simple voltage source, switch, and resistor circuit:



Note how the capacitor alternately functions as a source and as a load, depending on what it's connected to. When connected to a source of voltage, the capacitor absorbs (stores) energy in the form of an electric field between its plates. Current flows through the voltage source in the same direction as though it were powering a load (e.g. a resistor). When the capacitor's voltage equals the source voltage, current stops in the circuit. Flipping the switch to the "discharge" position connects the capacitor to a resistor, where it discharges its store of energy, acting as a source to the resistor's load.

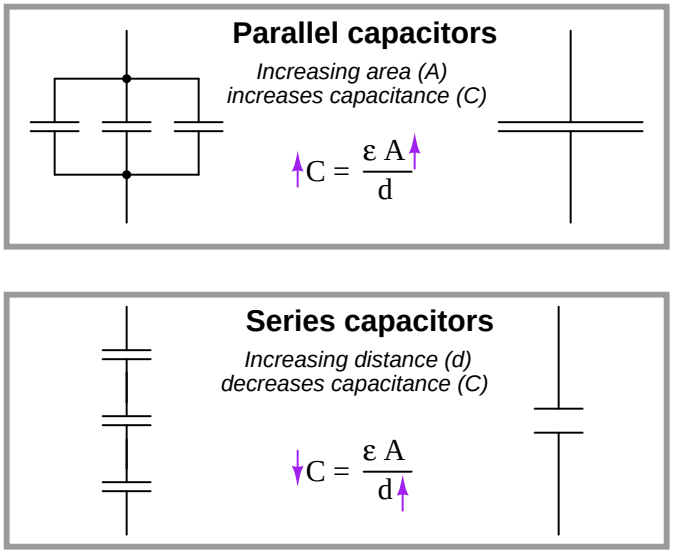
Note also how the capacitor maintains the same polarity of voltage during all three steps. Since energy is stored within an *electric field* inside the capacitor, and electric fields are a function of *voltage*, the stored energy manifests itself as a voltage with an unchanging polarity. The Law of Energy Conservation – which states energy cannot spontaneously vanish or appear but must be accounted for – tells us a capacitor's voltage cannot spontaneously vanish, appear, or change polarity. In other words, *voltage is the conserved quantity in a capacitor as it transitions between acting as a source and acting as a load.* Current through a capacitor, however, switches direction depending on whether the capacitor is charging (acting as a load) or discharging (acting as a source).

Capacitance adds when capacitors are connected in parallel. It diminishes when capacitors are connected in series:

$$C_{parallel} = C_1 + C_2 + \cdots C_n \qquad\qquad C_{series} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}}$$

Recall that capacitance stores energy in the form of an electric field, as a function of the voltage applied to it. It we wish to increase capacitance by connecting multiple capacitors to each other, we need to do that in such a way that each capacitor receives the same (total) applied voltage so that each additional amount of capacitance included in the network will contribute a proportional amount of energy storage to the network. We know voltage is guaranteed to be equal only among *parallel*-connected components. If we were to connect multiple capacitors in series with one another, their individual voltages would be some fraction of the total voltage (series voltages always adding to equal the total voltage), thus diminishing the energy stored in each capacitor and similarly diminishing the total capacitance.

Another way to comprehend why capacitance increases in parallel and diminishes in series is to consider a network of capacitors as one equivalent capacitor in terms of aggregate plate area and separation distance. Examining the directions of change to the corresponding variables in the permittivity/area/distance formula for capacitance, we see how parallel-connected capacitances must add while series-connected capacitances must diminish:



The relationship between voltage and current for a capacitor is as follows:

$$I = C\frac{dV}{dt}$$

As such, capacitors oppose changes in voltage over time by passing a current. This behavior makes capacitors useful for stabilizing voltage in DC circuits. One way to think of a capacitor in a DC circuit is as a *temporary voltage source*, always "wanting" to maintain voltage across its terminals as a function of the energy stored within its electric field.

The amount of potential energy (E_p, in units of joules) stored by a capacitor may be determined by altering the voltage/current/capacitance equation to express power ($P = IV$) and then applying some calculus (recall that power is defined as the time-derivative of work or energy, $P = \frac{dW}{dt} = \frac{dE}{dt}$):

$$I = C\frac{dV}{dt}$$

$$P = IV = CV\frac{dV}{dt}$$

$$\frac{dE_p}{dt} = CV\frac{dV}{dt}$$

$$\frac{dE_p}{dt}\,dt = CV\,dV$$

$$\int \frac{dE_p}{dt}\,dt = \int CV\,dV$$

$$\int dE_p = C\int V\,dV$$

$$E_p = \frac{1}{2}CV^2$$

In an AC circuit, the amount of capacitive reactance (X_C) offered by a capacitor is inversely proportional to both capacitance and frequency:

$$X_C = \frac{1}{2\pi fC}$$

This means an AC signal finds it "easier" to pass through a capacitor (i.e. less ohms of reactance) at higher frequencies than at lower frequencies.

4.13 Inductors

Any conductor possesses a characteristic called *inductance*: the ability to store energy in the form of a magnetic field created by an electric current flowing through that conductor. Inductance is symbolized by the capital letter L and is measured in the unit of the *Henry* (H).

Inductors are devices expressly designed and manufactured to possess inductance. They are typically constructed of a wire coil wound around a ferromagnetic[17] core material. Inductors have current ratings as well as inductance ratings. Due to the effect of *magnetic saturation*, inductance tends to decrease as current approaches the rated maximum value in an iron-core inductor. Here are some schematic symbols for inductors:

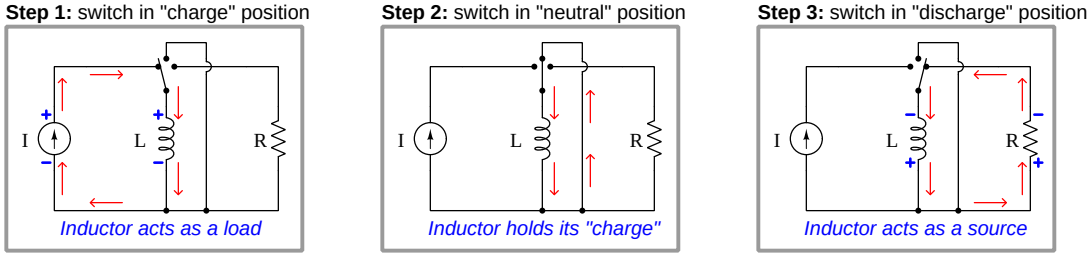


An inductor's inductance is related to the magnetic permeability of the core material (μ), the number of turns in the wire coil (N), the cross-sectional area of the coil (A), and the length of the coil (l):

$$L = \frac{\mu N^2 A}{l}$$

[17] The term "ferrous" refers to any substance containing the element *iron*. Steel is one such substance, being a combination of iron and carbon.

Inductance is a non-dissipative quantity. Unlike resistance, a pure inductance does not dissipate energy in the form of heat; rather, it stores and releases energy from and to the rest of the circuit. We may illustrate the energy-storing behavior of an ideal[18] inductor by this simple current source, switch, and resistor circuit:



Note how the inductor alternately functions as a source and as a load, depending on what it's connected to. When connected to a source of current, the inductor absorbs (stores) energy in the form of a magnetic field within its core. Voltage is dropped across the current source in the same polarity as though it were powering a load (e.g. a resistor). When the inductor's current equals the rated current of the source, there will no longer be a voltage drop (assuming a perfect inductor with zero internal wire resistance). Flipping the switch to the "discharge" position connects the inductor to a resistor, where it discharges its store of energy, acting as a source to the resistor's load.

Note also how the inductor maintains the same direction of current. Since energy is stored within a *magnetic field* inside the inductor, and magnetic fields are a function of *current*, the stored energy manifests itself as a current with an unchanging direction. The Law of Energy Conservation – which states energy cannot spontaneously vanish or appear but must be accounted for – tells us an inductor's current cannot spontaneously vanish, appear, or change direction. In other words, *current is the conserved quantity in an inductor as it transitions between acting as a source and acting as a load.* Voltage across an inductor, however, switches polarity depending on whether the inductor is charging (acting as a load) or discharging (acting as a source).

[18] Unlike the charge/hold/discharge capacitor circuit, this inductor demonstration circuit would not function quite as well in real life. Real inductors contain substantial amounts of electrical resistance (R) in addition to inductance (L), which means real inductors have an inherent capacity to dissipate their own store of energy. If a real inductor were placed in a circuit such as this, it would not maintain its store of energy indefinitely in the switch's "neutral" position as a capacitor would. Realistically, the inductor's energy would likely dissipate in a matter of milliseconds following the switch to the "neutral" position.

Inductance adds when inductors are connected in series. It diminishes when inductors are connected in parallel:

$$L_{series} = L_1 + L_2 + \cdots L_n \qquad\qquad L_{parallel} = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_n}}$$

Recall that inductance stores energy in the form of a magnetic electric field, as a function of the electric current flowing through it. It we wish to increase inductance by connecting multiple inductors to each other, we need to do that in such a way that each inductor receives the same (total) current so that each additional amount of inductance included in the network will contribute a proportional amount of energy storage to the network. We know current is guaranteed to be equal only among *series*-connected components. If we were to connect multiple inductors in parallel with one another, their individual currents would be some fraction of the total current (parallel currents always adding to equal the total current), thus diminishing the energy stored in each inductor and similarly diminishing the total inductance.

The relationship between voltage and current for an inductor is as follows:

$$V = L\frac{dI}{dt}$$

As such, inductors oppose changes in current over time by dropping a voltage. This behavior makes inductors useful for stabilizing current in DC circuits. One way to think of an inductor in a DC circuit is as a *temporary current source*, always "wanting" to maintain current through its coil as a function of the energy stored within its magnetic field.

The amount of potential energy (E_p, in units of joules) stored by an inductor may be determined by altering the voltage/current/inductance equation to express power ($P = IV$) and then applying some calculus (recall that power is defined as the time-derivative of work or energy, $P = \frac{dW}{dt} = \frac{dE}{dt}$):

$$V = L\frac{dI}{dt}$$

$$P = IV = LI\frac{dI}{dt}$$

$$\frac{dE_p}{dt} = LI\frac{dI}{dt}$$

$$\frac{dE_p}{dt}\,dt = LI\,dI$$

$$\int \frac{dE_p}{dt}\,dt = \int LI\,dI$$

$$\int dE_p = L\int I\,dI$$

$$E_p = \frac{1}{2}LI^2$$

In an AC circuit, the amount of inductive reactance (X_L) offered by an inductor is directly proportional to both inductance and frequency:

$$X_L = 2\pi f L$$

This means an AC signal finds it "harder" to pass through an inductor (i.e. more ohms of reactance) at higher frequencies than at lower frequencies.

References

Boylestad, Robert L., *Introductory Circuit Analysis*, 9th Edition, Prentice Hall, Upper Saddle River, NJ, 2000.